

SOTER: A Generative Time-Series Foundation Model for Wearable Human Physiological Signals

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Abstract

Time-series foundation models have demonstrated strong cross-domain transfer, yet their common architectural assumptions remain poorly aligned with wearable physiological signals, which are multichannel, irregularly sampled, noisy, and governed by coupled continuous-time dynamics spanning distinct spectral scales. We present **SOTER**, a generative foundation model for wearable physiological time series that unifies cross-channel coupling, spectrum-guided expert specialization, and continuous-time latent evolution within a single pre-training framework. SOTER combines a spatial feature-aware backbone that models inter-signal dependencies, a power spectral density (PSD)-guided mixture-of-experts layer that routes representations to experts associated with fixed spectral bands through an inspectable, non-learned rule, and a neural controlled differential equation decoder that supports prediction and imputation at arbitrary timestamps. We pre-train SOTER on 226 billion time points from five public physiological datasets and evaluate the same pre-trained model across out-of-distribution zero-shot forecasting, frozen-encoder linear-probe classification, and continuous-time imputation on wearable benchmarks. SOTER achieves the best RMSE on 4 of 6 datasets and the best MAE on 5 of 6 in zero-shot forecasting, the highest average Macro-AUROC in classification, and the lowest imputation error on all six datasets at 75% missingness. It further remains robust to additive acquisition noise, matching or surpassing baselines evaluated on clean inputs even under the strongest corruption. These results indicate that domain-specialized foundation models for wearable physiology benefit from jointly modeling channel structure, spectral scale, and continuous-time dynamics¹.

1 Introduction

Wearable sensors enable continuous and long-term monitoring of human physiological signals. Such signals—including electroencephalography (EEG) reflecting brain activity, photoplethysmography (PPG), electrodermal activity (EDA), and blood volume pulse

(BVP)—provide complementary measurements of human physiological states [11, 15]. Capturing the underlying patterns within these signals through large-scale representation learning can support downstream tasks such as emotion recognition and seizure detection [16, 41], enabling earlier anomaly warning and more timely clinical action [25, 50]. In practice, however, physiological data collected across populations, devices, and institutions often exhibit substantial distributional differences [58], while privacy regulations such as GDPR further constrain cross-region data aggregation [51]. These conditions motivate a domain-specialized time-series foundation model that can be pre-trained on de-identified public data and transferred across datasets and tasks with limited downstream adaptation.

Time-series foundation models (TSFMs) provide a basis for such unified modeling. Generalist TSFMs have advanced through multimodal inputs, flexible generative objectives, and sparse mixture-of-experts (MoE) architectures [34, 53, 57], but they are primarily designed around generic time-series assumptions. Most retain channel independent processing [42], employ learned expert routing whose specialization emerges implicitly during optimization, or operate on discrete temporal grids. Recent medical and wearable TSFMs have begun to incorporate domain-specific designs: PaPaGei pre-trains unified representations for the PPG modality [43]; MIRA incorporates neural ordinary differential equations (Neural ODEs) [7] and continuous-time rotary position encoding for irregular observations [27]; and NormWear introduces channel-aware attention through a shared liaison token [36]. Nevertheless, existing approaches address only individual aspects of physiological modeling and do not jointly capture channel coupling, spectral specialization, and continuous-time generation within a unified pre-training framework. This gap reflects a broader mismatch between the assumptions of existing time-series models and the intrinsic structure of wearable physiological signals.

These physiological characteristics distinguish wearable signals from generic time series in three key ways. First, physiological channels are not independent signals; together, they provide complementary observations of a shared underlying state. State-dependent interactions exist among physiological systems such as the heart,

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¹Source code is available at <https://github.com/SII-fkchen/SOTER>

respiratory system, and nervous system [5, 37]. Processing channels independently may therefore discard coordinated patterns that cannot be recovered from any single signal alone. Second, physiological dynamics span heterogeneous temporal and spectral scales. These scales interact with physiological states: heart rate, blood pressure, and respiration exhibit cross-scale dependencies, while different physiological rhythms show characteristic spectral patterns under varying states [1, 29]. Uniform transformations across these regimes may obscure scale-specific structures and force shared parameters to model substantially different dynamics. Third, physiological processes evolve continuously, whereas wearable devices capture them at discrete, often irregular, and incomplete timestamps. Such acquisition patterns lead to missing observations and reduced signal reliability in real-world wearable monitoring [8, 20]. Models restricted to fixed temporal grids cannot naturally query latent physiological states at arbitrary timestamps, limiting both future forecasting and causal imputation. Together, these properties motivate three requirements for a physiological foundation model: cross-channel coupling, spectrum-conditioned capacity allocation, and continuous-time state evolution.

To meet these requirements, we propose SOTER, a generative time-series foundation model for wearable human physiological signals. SOTER employs a spatial feature-aware backbone that preserves channel-specific temporal modeling in its lower layers and restores cross-channel interaction over high-level representations. It further introduces a power spectral density (PSD)-guided mixture-of-experts layer, whose non-learned routing rule associates experts with fixed spectral bands, providing frequency-grounded and inspectable specialization without an auxiliary routing loss. Finally, a terminal decoder based on neural controlled differential equations (Neural CDEs) and Neural ODEs evolves the latent physiological state to arbitrary query timestamps, thereby supporting both future forecasting and causal imputation [7, 24]. These components are not independent add-ons: they respectively model the structure, spectral scale, and temporal evolution of the same latent physiological process within a unified generative objective. Our main contributions are as follows:

- We introduce SOTER, a domain-specialized generative time-series foundation model for wearable physiological signals, pre-trained on 226 billion time points from five public physiological datasets. SOTER enables unified transfer across zero-shot forecasting, frozen-representation classification, and continuous-time imputation from a single pre-trained backbone.
- We develop a physiology-aware architecture that integrates cross-channel coupling, PSD-guided expert specialization, and continuous-time latent evolution, aligning model capacity with the structural, spectral, and temporal properties of physiological signals.
- Extensive experiments across six out-of-distribution wearable datasets demonstrate that SOTER achieves consistent improvements in zero-shot forecasting, frozen representation evaluation, and continuous-time imputation.
- We further analyze the source of SOTER’s transferability through corpus-matched pre-training comparisons and

evaluations under noisy observations, limited supervision, and different pre-training exposures.

2 Related Work

2.1 Generalist Time-Series Foundation Models

Generalist TSFMs are pre-trained on large and heterogeneous corpora and have demonstrated strong zero-shot transfer across forecasting domains [14, 31, 33]. Recent architectures span multimodal inputs that combine rendered time-series images, text prompts, and raw signals [57], flow-matching objectives for flexible generative modeling [34], group attention for information sharing among related series [2], and sparse mixture-of-experts architectures for scalable capacity [32, 53]. Other representative families, including TimesFM [10], Moirai [30], MOMENT [13], TiRex [4], TTM [12], and Lag-Llama [45], further explore diverse forecasting and representation learning designs. Although several of these models support multivariate inputs or information sharing across related series, their dominant assumptions remain generic: observations are represented on discrete temporal grids, channel interactions are not designed around coupled physiological systems, and expert specialization, when present, is typically determined implicitly through learned routing. They therefore provide strong general-purpose baselines, but do not directly address the combined structural requirements of wearable physiological signals.

2.2 Medical Time-Series Models

Deep learning has advanced medical time-series analysis across forecasting, phenotyping, monitoring, diagnosis, and public-health surveillance [6, 17, 26, 28, 38, 44, 48, 59, 60]. Most conventional approaches, however, are optimized for a specific task, signal type, or clinical scenario, limiting their reuse across heterogeneous physiological datasets and downstream objectives. Recent medical and wearable TSFMs have begun to improve cross-domain transfer. PaPaGei [43] and Pulse-PPG [49] learn representations specialized for PPG; MIRA [27] incorporates Neural ODEs and continuous-time rotary position encoding to accommodate irregular observations; and NormWear [36] introduces a shared liaison token to capture relationships among wearable sensor channels.

These models address important subsets of the physiological modeling problem, but their capabilities remain complementary rather than unified. Modality-specific models do not provide generative modeling across heterogeneous physiological signals; continuous-time medical models do not explicitly allocate capacity according to spectral content; and channel-aware representation models are not designed to support arbitrary-time forecasting and causal imputation through a shared generative decoder. SOTER instead unifies high-level cross-channel coupling, spectrum-conditioned expert specialization, and continuous-time generative decoding within a single pre-trained time-series model evaluated across forecasting, classification, and imputation.

3 Methodology

3.1 Problem Statement

We consider a collection of wearable physiological recordings, each comprising n channels observed at a shared, possibly irregularly

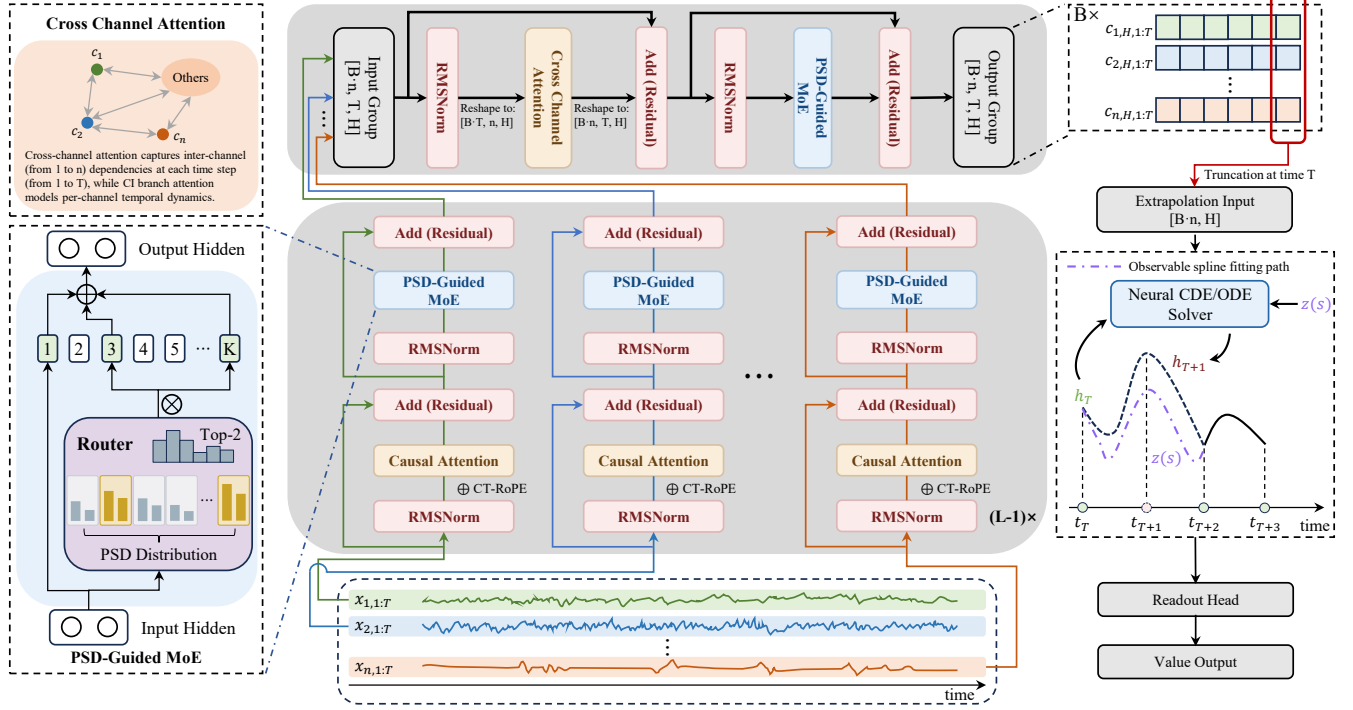


Figure 1: Overview of SOTER. The backbone consists of $L-1$ CI layers followed by one CD layer. The CI layers model the temporal dynamics of each channel independently, whereas the CD layer captures cross-channel dependencies over high-level representations. A terminal decoder based on Neural CDEs advances the latent state h_T to an arbitrary query time using a control path $z(s)$ interpolated from the causal observation prefix. When the query lies beyond the observed interval and no control is available, the decoder reduces to a Neural ODE.

spaced sequence of timestamps $t_{1:T} = (t_1, \dots, t_T)$; the value of channel $c \in \{1, \dots, n\}$ at timestamp t_i is a scalar $x_{c,i}$. Given an observed history of wearable physiological recordings, SOTER is pre-trained in a generative fashion to predict the value at the next queried timestamp. An overview of the architecture is shown in Figure 1.

3.2 SOTER Architecture

3.2.1 Continuous-Time Input Encoding. Each scalar observation $x_{c,i}$ is first lifted to an H -dimensional embedding through a gated projection:

$$h_{c,i}^0 = (\text{SiLU}(W_g x_{c,i})) \odot (W_e x_{c,i}), \quad (1)$$

where W_g and W_e map the single input channel to the hidden width, and $\odot$ denotes the elementwise product. To avoid premature cross-channel leakage, each channel is first treated as an independent univariate sequence in the lower layers [42]. Because physiological signals are frequently sampled at irregular intervals, discrete positional indices poorly reflect their temporal geometry; SOTER therefore employs continuous-time rotary positional encoding (CT-RoPE) [27], which rotates queries and keys by an angle proportional to the absolute timestamp rather than the token position.

3.2.2 Spatial Feature-Aware Backbone. The backbone stacks $L=6$ pre-norm blocks, each employing multi-head attention with a per-head dimension of d ; the attention operator is layer-dependent,

with the first $L-1$ layers being channel-independent (CI) and a single high-level layer being channel-dependent (CD). Within each backbone block, the feed-forward sublayer is implemented as the PSD-guided mixture-of-experts module, while the attention sublayer follows the CI/CD design described above. In a CI layer, attention is applied independently to each channel sequence, whose hidden states form $U = [h_1; \dots; h_T] \in \mathbb{R}^{T \times H}$, with $h_i \in \mathbb{R}^H$ denoting the representation at timestamp t_i , using causal self-attention modulated by CT-RoPE:

$$\mathcal{A}_{ij}^{\text{CI}} = h_i^\top W_q \mathbf{R}_{\Theta, t_i - t_j} W_k^\top h_j, \quad (2)$$

$$\text{Attention}^{\text{CI}}(U) = \text{Softmax}\left(\frac{\text{Mask}(\mathcal{A}^{\text{CI}})}{\sqrt{d}}\right) U W_v.$$

Here, $W_q, W_k, W_v \in \mathbb{R}^{H \times d}$ are the query, key, and value projections, $\mathbf{R}_{\Theta, t} \in \mathbb{R}^{d \times d}$ is the CT-RoPE rotary matrix whose rotation angle is determined by the absolute timestamp t , and $\text{Mask}(\cdot)$ is the causal mask that forbids attending to future timestamps. Equations (2) and (3) are written for a single head; the standard M -head extension ($Md = H$, with outputs mixed by $W_o \in \mathbb{R}^{Md \times H}$) is omitted.

In the CD layer, cross-channel coupling is restored, which is essential because the human body behaves as a strongly coupled dynamical system [5]. After channel-independent temporal modeling, representations are regrouped at each timestamp to restore

cross-channel interaction. For a given timestamp, the channel representations are denoted as $\tilde{U} = [\tilde{h}_1; \dots; \tilde{h}_n] \in \mathbb{R}^{n \times H}$, where $\tilde{h}_a$ represents the a -th channel. Full non-causal attention is then computed over the n channel tokens at each timestamp:

$$\mathcal{A}_{ab}^{\text{CD}} = \tilde{h}_a^\top W_{\tilde{q}} W_{\tilde{k}}^\top \tilde{h}_b, \quad (3)$$

$$\text{Attention}^{\text{CD}}(\tilde{U}) = \text{Softmax}\left(\frac{\mathcal{A}^{\text{CD}}}{\sqrt{d}}\right) \tilde{U} W_{\tilde{v}}.$$

Here, $a, b \in \{1, \dots, n\}$ index the channels, and $W_{\tilde{q}}, W_{\tilde{k}}, W_{\tilde{v}} \in \mathbb{R}^{H \times d}$ are the query, key, and value projections of the CD layer. No positional encoding or causal mask is applied along the channel axis, since channels carry no intrinsic ordering. Confining cross-channel attention to a single top layer keeps the $\mathcal{O}(n^2)$ cost over channels to one block, while letting it operate over refined, rather than shallow, temporal representations. When $n=1$, this layer falls back to standard channel-independent temporal attention, so SOTER transfers across recordings with different channel counts.

3.2.3 PSD-Guided Mixture-of-Experts. Prevailing MoE-based foundation models route tokens with a learned gate regularized by an auxiliary load-balancing loss [27, 53], so that expert assignment follows no explicit criterion and emerges only from training. SOTER instead derives routing directly from the spectral content of the latent representation, using a PSD-inspired statistic to provide an explicit frequency-aware criterion for expert assignment. For each latent trajectory, we maintain a strictly causal, N -point prefix discrete Fourier transform (DFT) [9] over the latent temporal index. Here, t denotes the current latent position and τ denotes the historical index within the causal prefix. We then average its magnitude over the hidden dimension:

$$X_t^{(d)}(k) = \sum_{\tau \leq t} h_\tau^{(d)} e^{-j2\pi k\tau/N}, p_t(k) = \frac{1}{H} \sum_{d=1}^H |X_t^{(d)}(k)|. \quad (4)$$

The transform is taken over latent temporal indices, i.e., on the normalized frequency axis of the latent trajectory. Absolute and possibly irregular timing is handled separately by CT-RoPE and the CDE decoder, so the router needs only a sampling-rate-invariant spectral-shape statistic rather than a physical-frequency spectrum, which keeps expert assignment consistent across the heterogeneous pre-training corpora.

The $\lfloor N/2 \rfloor + 1$ frequency bins are partitioned into E contiguous bands $\{\mathcal{B}_e\}$. The routing logit of expert e is the aggregated spectral magnitude within $\mathcal{B}_e$, from which the top-2 experts are selected and weighted:

$$g_{t,e} = \sum_{k \in \mathcal{B}_e} p_t(k), \quad (5)$$

$$r_{t,e} = \frac{\exp(g_{t,e})}{\sum_{e'=1}^E \exp(g_{t,e'})} \mathbf{1}[e \in \text{TopK}(g_t, 2)].$$

The band-wise statistic $g_{t,e}$ aggregates spectral magnitudes within $\mathcal{B}_e$; magnitude accumulation avoids excessive amplification of dynamic ranges across bands. Because the routing statistic is computed without gradients and the prefix transform depends only on past tokens, the assignment is deterministic, strictly causal, and free of any auxiliary balancing loss, so each expert is tied to a fixed spectral band by construction.

A shared expert acting on the sequence-averaged representation is finally added as a global residual:

$$\text{MoE}(h_t) = \mathcal{E}_{\text{sh}}\left(\frac{1}{T} \sum_{\tau} h_\tau\right) + \sum_{e \in \text{TopK}(g_t, 2)} r_{t,e} \mathcal{E}_e(h_t). \quad (6)$$

This shared branch is a sequence-level residual rather than a per-token causal path: it adds the mean over the window's T tokens to every position. Since the prediction target t_{T+1} lies strictly after the entire window, every token entering this aggregate belongs to the causal prefix of the target, so the shared branch introduces no future leakage with respect to the prediction and remains consistent with the strictly causal prefix-DFT routing and causal attention used throughout.

3.2.4 Neural CDE Extrapolation Decoder. Autoregressive architectures generate under causal masking and cannot access the target timestamp at inference. To extrapolate to arbitrary timestamps, SOTER takes the state h_T at the last observed time t_T of each channel and evolves it to the target time t_{T+1} through a controlled continuous-time field. Unlike a Neural ODE [7], which depends only on time and state, the vector field of SOTER additionally receives an external control path $z(s)$:

$$\frac{dh(s)}{ds} = f_\theta(s, h(s), z(s)), h(t_T) = h_T, s \in [t_T, t_{T+1}]. \quad (7)$$

Here, f_θ is a time-augmented multilayer perceptron (MLP) taking the concatenation $[s; h(s); z(s)]$ as input. The target state follows from numerical integration with an adaptive Dormand–Prince 5(4) solver (dopri5):

$$h(t_{T+1}) = h_T + \int_{t_T}^{t_{T+1}} f_\theta(s, h(s), z(s)) ds, \quad (8)$$

from which a linear readout head produces the prediction $\hat{x}_{T+1} = W_o h(t_{T+1})$.

During inference, the same continuous-time decoder supports two querying regimes through the control path $z(s)$. When the target timestamp lies within the span of observations, the decoder operates as a Neural CDE-style controlled regime [24]: the observed points are interpolated into $z(s)$ by natural cubic splines, and the latent dynamics are steered by the observable signal. Although a natural cubic spline is a global interpolant, the spline for each queried point is fitted only on observations strictly before the prediction origin—later samples are physically excluded from the input rather than down-weighted by masking—so the control path remains strictly causal, consistent with the online streaming imputation setting. When no control is available beyond the observation span, we set $z(s) \equiv 0$, and the decoder reduces exactly to a time-augmented Neural ODE [7]. We provide more details in Appendix D. A single continuous-time decoder thus unifies strictly causal extrapolation and observation-guided interpolation, which underpins the transferability of a single frozen SOTER across both forecasting and imputation.

3.3 Model Training

SOTER is pre-trained on the next-timestamp prediction task over a wearable physiological corpus of approximately 226 billion time points, assembled from five publicly available and de-identified datasets: MIMIC-III Waveform [22, 39], Sleep-EDF [23], PTB-XL [55],

Table 1: Zero-shot forecasting on six out-of-distribution wearable datasets. Values are reported as $\text{mean}_{\pm\text{std}}$ over four input/output window pairs (48/24, 72/36, 96/48, 128/64); Timer is evaluated on the latter two only, due to its native configuration. Entries are rescaled by 10^2 . Paras denotes parameters activated per inference step (millions), and SC and MC denote single- and multi-channel datasets. The best result is indicated by bold style with an asterisk (*), while the second-best result is underlined.

Method	Paras	Source	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)		IEEE PPG (MC)		MMASH (MC)		WDD (MC)	
			RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Timer	84.14M	ICML'24	8.90 $\pm$ 1.26	4.85 $\pm$ 0.52	6.11 $\pm$ 0.25	3.64 $\pm$ 0.01	15.60 $\pm$ 0.01	11.92 $\pm$ 0.01	12.92 $\pm$ 0.54	8.71 $\pm$ 0.39	2.91 $\pm$ 0.28	2.19 $\pm$ 0.13	3.32 $\pm$ 0.10	2.70 $\pm$ 0.09
TTM	0.805M	NIPS'24	5.05 $\pm$ 0.84	2.82 $\pm$ 0.52	4.14 $\pm$ 0.11	2.06 $\pm$ 0.10	9.60 $\pm$ 0.09	7.43 $\pm$ 0.21	9.15 $\pm$ 0.20	6.06 $\pm$ 0.24	1.48 $\pm$ 0.44	0.94 $\pm$ 0.22	2.07 $\pm$ 0.24	1.60 $\pm$ 0.25
Lag-Llama	2.451M	ARXIV'23	19.96 $\pm$ 0.32	15.63 $\pm$ 0.13	48.93 $\pm$ 0.09	48.64 $\pm$ 0.09	51.02 $\pm$ 0.37	46.92 $\pm$ 0.40	51.75 $\pm$ 0.09	50.69 $\pm$ 0.10	18.37 $\pm$ 0.09	17.59 $\pm$ 0.07	43.75 $\pm$ 0.18	41.45 $\pm$ 0.01
TiRex	35.29M	NIPS'25	5.22 $\pm$ 1.25	2.49 $\pm$ 0.55	4.15 $\pm$ 0.32	2.02 $\pm$ 0.12	5.82 $\pm$ 0.52	4.05 $\pm$ 0.51	8.93 $\pm$ 0.36	5.35 $\pm$ 0.27	0.37 $\pm$ 0.25	0.16 $\pm$ 0.13	<u>0.98</u> $\pm$ 0.07	0.77 $\pm$ 0.09
MOMENT-Large	341.3M		6.33 $\pm$ 0.40	3.36 $\pm$ 0.23	4.23 $\pm$ 0.11	2.21 $\pm$ 0.05	9.74 $\pm$ 1.17	7.90 $\pm$ 0.92	10.60 $\pm$ 0.72	6.61 $\pm$ 0.50	0.69 $\pm$ 0.31	0.31 $\pm$ 0.14	1.28 $\pm$ 0.24	0.70 $\pm$ 0.16
MOMENT-Base	109.6M	ICML'24	6.94 $\pm$ 0.31	3.82 $\pm$ 0.34	4.80 $\pm$ 0.30	2.52 $\pm$ 0.07	9.85 $\pm$ 1.16	7.76 $\pm$ 0.96	10.91 $\pm$ 0.82	6.76 $\pm$ 0.55	0.65 $\pm$ 0.30	0.30 $\pm$ 0.13	1.33 $\pm$ 0.28	0.74 $\pm$ 0.16
MOMENT-Small	35.34M		6.59 $\pm$ 0.34	3.47 $\pm$ 0.34	4.38 $\pm$ 0.10	2.30 $\pm$ 0.05	9.47 $\pm$ 1.22	7.55 $\pm$ 0.98	11.06 $\pm$ 0.75	6.84 $\pm$ 0.51	0.71 $\pm$ 0.30	0.32 $\pm$ 0.15	1.28 $\pm$ 0.27	0.69 $\pm$ 0.15
MoiraiMoE-Base	935.1M		7.00 $\pm$ 0.81	3.43 $\pm$ 0.71	5.47 $\pm$ 0.55	2.50 $\pm$ 0.23	13.18 $\pm$ 0.71	9.25 $\pm$ 0.58	11.68 $\pm$ 0.74	7.06 $\pm$ 0.20	0.79 $\pm$ 0.52	0.30 $\pm$ 0.19	1.27 $\pm$ 0.10	1.05 $\pm$ 0.04
MoiraiMoE-Small	117.0M	ICML'25	7.30 $\pm$ 0.80	3.65 $\pm$ 0.67	5.25 $\pm$ 0.38	2.41 $\pm$ 0.16	13.43 $\pm$ 0.68	9.60 $\pm$ 0.30	12.02 $\pm$ 0.70	7.16 $\pm$ 0.09	0.77 $\pm$ 0.44	0.30 $\pm$ 0.16	1.25 $\pm$ 0.12	1.04 $\pm$ 0.07
Moirai2.0-Small	11.39M	Salesforce'26	5.33 $\pm$ 0.28	2.65 $\pm$ 0.40	4.59 $\pm$ 0.47	2.16 $\pm$ 0.17	8.12 $\pm$ 0.44	4.23 $\pm$ 0.13	9.19 $\pm$ 0.81	5.61 $\pm$ 0.38	0.32 $\pm$ 0.27	0.12 $\pm$ 0.10	1.07 $\pm$ 0.13	0.89 $\pm$ 0.06
TimesFM 1.0	203.6M	ICML'24	6.04 $\pm$ 1.00	2.78 $\pm$ 0.37	4.38 $\pm$ 0.09	2.14 $\pm$ 0.06	10.74 $\pm$ 1.05	7.94 $\pm$ 0.61	8.60 $\pm$ 0.62	5.30 $\pm$ 0.09	0.33 $\pm$ 0.05	0.11 $\pm$ 0.03	1.03 $\pm$ 0.14	0.72 $\pm$ 0.08
TimesFM 2.0	498.8M	Google'25	6.46 $\pm$ 0.79	3.07 $\pm$ 0.37	4.42 $\pm$ 0.37	2.09 $\pm$ 0.11	11.02 $\pm$ 0.76	8.09 $\pm$ 0.34	9.24 $\pm$ 0.88	5.67 $\pm$ 0.13	<u>0.23</u> $\pm$ 0.09	<u>0.08</u> $\pm$ 0.04	1.00 $\pm$ 0.12	0.69 $\pm$ 0.03
TimesFM 2.5	231.3M		5.55 $\pm$ 0.54	2.48 $\pm$ 0.27	4.11 $\pm$ 0.32	1.97 $\pm$ 0.11	10.87 $\pm$ 1.13	7.86 $\pm$ 0.76	8.79 $\pm$ 0.77	5.35 $\pm$ 0.18	0.08 $\pm$ 0.04	0.03 $\pm$ 0.02	1.05 $\pm$ 0.16	0.81 $\pm$ 0.10
TimeMoE-Large	453.2M		<u>1.68</u> $\pm$ 0.67	<u>1.16</u> $\pm$ 0.32	<u>1.82</u> $\pm$ 0.11	<u>1.00</u> $\pm$ 0.03	<u>3.10</u> $\pm$ 0.09	<u>1.95</u> $\pm$ 0.07	<u>1.77</u> $\pm$ 0.07	<u>1.08</u> $\pm$ 0.01	3.06 $\pm$ 0.24	2.61 $\pm$ 0.20	1.57 $\pm$ 0.07	1.21 $\pm$ 0.03
TimeMoE-Base	113.3M	ICLR'25	1.85 $\pm$ 0.33	1.30 $\pm$ 0.22	2.15 $\pm$ 0.17	1.39 $\pm$ 0.21	3.51 $\pm$ 0.13	2.22 $\pm$ 0.06	<u>1.79</u> $\pm$ 0.07	1.14 $\pm$ 0.02	3.31 $\pm$ 0.23	2.65 $\pm$ 0.16	2.16 $\pm$ 0.13	1.67 $\pm$ 0.07
Chronos2-Base	119.5M		5.65 $\pm$ 1.01	2.49 $\pm$ 0.40	4.80 $\pm$ 1.19	2.11 $\pm$ 0.34	7.21 $\pm$ 0.70	3.88 $\pm$ 0.25	9.57 $\pm$ 1.16	5.79 $\pm$ 0.54	<u>0.23</u> $\pm$ 0.16	0.10 $\pm$ 0.05	1.09 $\pm$ 0.07	0.63 $\pm$ 0.15
Chronos2-Small	27.93M	Amazon'25	5.67 $\pm$ 0.30	2.73 $\pm$ 0.32	4.54 $\pm$ 0.80	2.11 $\pm$ 0.34	10.45 $\pm$ 1.52	7.17 $\pm$ 1.13	9.47 $\pm$ 1.05	5.77 $\pm$ 0.56	0.29 $\pm$ 0.17	0.11 $\pm$ 0.06	1.10 $\pm$ 0.10	0.64 $\pm$ 0.14
Sundial-Base	128.3M	ICML'25	5.75 $\pm$ 1.01	2.77 $\pm$ 0.33	4.12 $\pm$ 0.19	2.00 $\pm$ 0.05	10.75 $\pm$ 0.58	7.47 $\pm$ 1.49	8.40 $\pm$ 0.48	5.22 $\pm$ 0.07	1.14 $\pm$ 0.32	0.97 $\pm$ 0.25	1.38 $\pm$ 0.14	0.95 $\pm$ 0.14
Aurora	210.8M	ICLR'26	7.13 $\pm$ 1.04	3.97 $\pm$ 0.43	5.10 $\pm$ 0.22	2.61 $\pm$ 0.09	13.79 $\pm$ 1.75	10.69 $\pm$ 1.47	10.30 $\pm$ 0.45	6.52 $\pm$ 0.20	1.99 $\pm$ 0.27	0.81 $\pm$ 0.07	1.17 $\pm$ 0.17	0.62 $\pm$ 0.05
MIRA-Large	339.9M	NIPS'25	2.66 $\pm$ 1.56	1.85 $\pm$ 1.11	2.20 $\pm$ 0.14	1.54 $\pm$ 0.12	4.76 $\pm$ 0.16	3.73 $\pm$ 0.12	2.57 $\pm$ 0.06	2.04 $\pm$ 0.01	6.23 $\pm$ 0.24	5.49 $\pm$ 0.27	3.36 $\pm$ 0.03	3.00 $\pm$ 0.03
SOTER	20.29M	/	1.41 $\pm$ 0.15	0.95 $\pm$ 0.11	1.29 $\pm$ 0.22	0.62 $\pm$ 0.10	2.26 $\pm$ 0.12	1.17 $\pm$ 0.04	2.19 $\pm$ 0.19	1.05 $\pm$ 0.05	1.26 $\pm$ 0.26	0.89 $\pm$ 0.15	0.73 $\pm$ 0.10	0.60 $\pm$ 0.09

56], WESAD [52], and Chapman-ECG [63, 64]. The training objective is a masked Huber loss [21], chosen for its robustness to the motion and contact artifacts common in wearable acquisition. Training alternates per step, with equal probability, between two decoding regimes that share the same next-timestamp target. In the full-history regime, the decoder extrapolates from fully observed inputs as a pure Neural ODE [7]; in the missingness regime, we randomly mask 15% of historical observations while preserving their timestamps. The masked values are zero-filled for backbone processing, while the remaining observations construct the spline control path, enabling a Neural CDE-style continuous-time decoder driven by observed signals [24]. Alternating between them jointly activates both decoder modes and additionally provides robustness pre-training under corrupted observations. We train SOTER for one epoch with a maximum sequence length of 512 on 8 $\times$ NVIDIA H200 GPUs, taking approximately 312 hours. The model comprises 49.79M parameters, of which 20.29M are activated per inference step under top-2 routing; key hyperparameters are provided in the Appendix B.

4 Experiments and Analysis

4.1 Experimental Setup

We evaluate SOTER on six publicly available and de-identified wearable physiological datasets: HeartRate [54] (single-channel), MIT-BIH [40], ScientiSST MOVE [3], IEEE PPG [62], MMASH [46, 47], and WDD [18, 19] (multi-channel). All datasets use a leave-one-subject-out (LOSO) split. Per-channel MinMax normalization is estimated exclusively from the training split, preventing distributional information from leaking from the test samples. We organize the compared methods into three groups. General-purpose TSFMs

include Aurora [57], Sundial [34], Chronos-2 [2], TimeMoE [53], TimesFM [10], Moirai 2.0 [30], Moirai-MoE [32], MOMENT [13], TiRex [4], TTM [12], Timer [35], and Lag-Llama [45]. Medical and wearable TSFMs include MIRA [27] and NormWear [36], while TFC [61] serves as a physiological representation-learning baseline. For each downstream task, we select the subset of methods compatible with its evaluation protocol: discriminative representation models for classification, and models with generative or reconstruction capabilities for forecasting and imputation, as detailed in the corresponding result tables.

All methods use their officially released checkpoints without updating the pre-trained backbones and share the same input windows, normalization procedure, and test splits as SOTER. Timer is evaluated on the two longest windows only because of its native configuration. All experiments are conducted on a single NVIDIA RTX 4090 GPU.

4.2 Zero-Shot Future Forecasting

We evaluate zero-shot future forecasting on six out-of-distribution (OOD) wearable physiological datasets, reporting RMSE and MAE in the normalized space (Table 1). SOTER achieves the best RMSE on four of the six datasets and the best MAE on five, while activating only 20.29M parameters per inference step.

MMASH is the only dataset on which SOTER achieves neither the best RMSE nor the best MAE. To provide additional context, its normalized permutation entropy (NPE) is 0.334, the lowest among the six evaluated datasets and substantially below the 0.878 measured on MIT-BIH. This difference indicates that MMASH exhibits markedly lower temporal complexity than the other evaluation datasets. We hypothesize that this property contributes to SOTER's

Table 2: Performance on the downstream classification task. We report Macro-F1 (%) and Macro-AUROC on three wearable datasets, together with the average Macro-AUROC across datasets. Mean and standard deviation over three runs. The best scores are highlighted by bold style, and the second-best scores are underlined.

Methods	WDD (3-Classes)		ScientiSST MOVE (7-Classes)		MMASH (6-Classes)		Average of Macro-AUROC
	Macro-F1	Macro-AUROC	Macro-F1	Macro-AUROC	Macro-F1	Macro-AUROC	
Random Baseline	32.90 $\pm$ 0.42	50.00 $\pm$ 0.00	7.40 $\pm$ 0.49	50.00 $\pm$ 0.00	13.61 $\pm$ 0.16	50.00 $\pm$ 0.00	50.00
TF-C	59.28 $\pm$ 0.86	77.66 $\pm$ 0.64	47.79 $\pm$ 2.35	91.89 $\pm$ 2.02	24.16 $\pm$ 0.86	<u>61.69</u> $\pm$ 0.65	<u>77.08</u>
MOMENT-Large	58.73 $\pm$ 1.16	78.45 $\pm$ 0.55	49.60 $\pm$ 0.55	88.55 $\pm$ 0.64	17.30 $\pm$ 0.44	53.87 $\pm$ 0.49	72.75
TimeMoE-Large	59.69 $\pm$ 0.49	76.95 $\pm$ 0.27	37.53 $\pm$ 1.17	85.94 $\pm$ 0.75	20.51 $\pm$ 0.16	59.29 $\pm$ 0.57	74.93
TiRex	58.57 $\pm$ 0.72	<u>79.56</u> $\pm$ 0.73	42.89 $\pm$ 0.95	89.52 $\pm$ 1.05	19.05 $\pm$ 0.75	55.25 $\pm$ 0.90	74.78
Sundial	58.88 $\pm$ 1.09	73.47 $\pm$ 1.21	48.69 $\pm$ 0.40	88.12 $\pm$ 0.81	20.54 $\pm$ 0.37	59.80 $\pm$ 0.52	73.80
Chronos2-Base	59.61 $\pm$ 0.69	78.90 $\pm$ 0.69	48.77 $\pm$ 0.42	<u>92.63</u> $\pm$ 0.50	17.92 $\pm$ 0.91	53.01 $\pm$ 0.49	74.85
NormWear	59.50 $\pm$ 0.72	75.85 $\pm$ 0.57	43.03 $\pm$ 0.16	91.92 $\pm$ 0.98	20.10 $\pm$ 0.78	61.22 $\pm$ 0.79	76.34
MIRA-Large	<u>59.95</u> $\pm$ 0.92	78.71 $\pm$ 1.13	<u>50.09</u> $\pm$ 1.22	90.07 $\pm$ 0.81	<u>23.52</u> $\pm$ 0.22	59.94 $\pm$ 1.20	76.24
SOTER	60.83 $\pm$ 0.82	81.70 $\pm$ 0.65	50.90 $\pm$ 0.36	95.62 $\pm$ 0.12	23.46 $\pm$ 0.68	65.04 $\pm$ 0.68	80.79

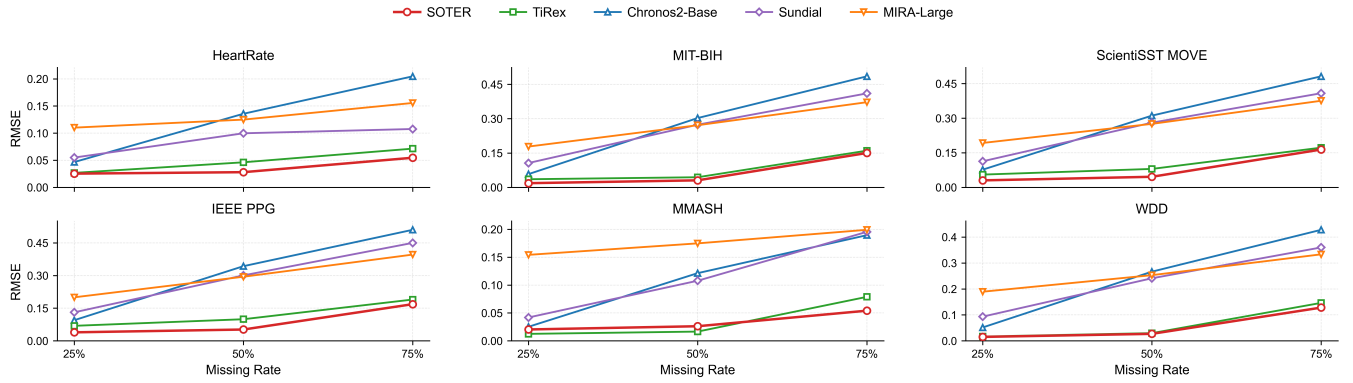


Figure 2: Continuous-time imputation under increasing missing rates. RMSE on six OOD wearable datasets at 25%, 50%, and 75% missing rates under the identical causal-prefix protocol. Values are means over three runs.

relative disadvantage on MMASH. Further dataset statistics and complexity analyses are provided in the Appendix A. Despite this exception, SOTER achieves the best performance on nine of the twelve dataset-metric combinations, demonstrating strong OOD transfer for zero-shot physiological forecasting.

4.3 Downstream Task Evaluation

We further evaluate the same frozen SOTER on classification and continuous-time imputation.

Classification. We adopt a frozen linear-probe protocol: the backbone remains fixed, and only a logistic-regression classifier is trained on top of its representations. This protocol provides a controlled measure of the discriminative information retained by the pre-trained model. Evaluation follows the LOSO split, and we report the mean and standard deviation over three runs. Because the evaluated datasets exhibit class imbalance, we report Macro-F1 and Macro-AUROC rather than accuracy, and use the average Macro-AUROC across datasets as the overall metric. As shown in Table 2, SOTER achieves the highest average Macro-AUROC of 80.79, outperforming the representation-learning baseline TF-C (77.08) and

the physiological foundation models NormWear (76.34) and MIRA (76.24). It also obtains the best Macro-AUROC on all three datasets. On MMASH, whose six-class distribution is the most imbalanced among the evaluated classification datasets, all methods achieve relatively low Macro-F1. SOTER trails TF-C on this metric but retains the highest Macro-AUROC, with all evaluated methods remaining above the random baseline in Macro-AUROC. Appendix I further shows that SOTER retains the strongest frozen linear-probe performance on ScientiSST MOVE across label budgets ranging from 1% to 100%.

Continuous-Time Imputation. Wearable signals are acquired and consumed as streams, where missing values must be estimated online using past observations alone. We therefore consider online causal imputation rather than offline bidirectional reconstruction. For each missing point, only its causal prefix is provided as context, and the CDE/ODE decoder evolves the latent state to the corresponding timestamp without write-back or autoregressive error accumulation. All models are evaluated under the same causal-prefix protocol, including identical missingness construction, evaluation points, and metrics. We consider missing rates of 25%, 50%, and

Table 3: Ablation study of SOTER on one single-channel and two multi-channel datasets. Entries are rescaled by 10^2 , as in Table 1. Bold indicates the best result in each column.

Method	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
SOTER	1.41	0.95	1.29	0.62	2.26	1.17
w/o CD Layer	1.43	0.98	2.14	1.68	3.78	2.15
w/o PSD-MoE	1.46	1.01	1.35	0.73	2.36	1.22
w/o Neural CDE	1.54	1.09	1.67	0.79	2.64	1.39
CD Layer@5th	1.42	0.97	1.47	0.82	2.32	1.25
CD Layer@4th	1.43	0.99	1.73	1.04	2.47	1.39
CD Layer@3rd	1.43	1.01	1.98	1.31	2.72	1.63

75%. Under this protocol, SOTER transfers to imputation zero-shot, without task-specific optimization. As shown in Figure 2, SOTER achieves the lowest RMSE on most datasets at each evaluated missing rate. At the highest missing rate of 75%, SOTER ranks first on all six datasets. TiRex outperforms SOTER on MMASH at the two lower missing rates. As missingness increases, Chronos-2, Sundial, and MIRA exhibit a more pronounced degradation, whereas SOTER and TiRex remain comparatively stable. This trend is consistent with the benefit of explicitly evolving the latent state toward arbitrary query timestamps under sparse causal observations. Full numerical results and standard deviations are provided in the Appendix G.

4.4 Ablation Analysis

We ablate the three core components of SOTER on the single-channel HeartRate dataset and the multi-channel MIT-BIH and ScientiSST MOVE datasets. In Table 3, w/o CD Layer replaces the channel-dependent layer with channel-independent processing; w/o PSD-MoE replaces the deterministic spectral router with a learned router trained using an auxiliary load-balancing loss; and w/o Neural CDE reduces the continuous-time decoder to a pure Neural ODE. Because the CDE control path is activated only in the missingness branch of pre-training, the final variant also removes this branch and therefore measures the combined contribution of the CDE mechanism and missingness-aware pre-training.

Removing any of the three components increases both RMSE and MAE on all evaluated datasets. Removing the CD layer has little effect on single-channel HeartRate but causes substantially larger degradation on the two multi-channel datasets, supporting its role in modeling cross-channel dependencies. Removing the Neural CDE also increases the error on HeartRate, showing that its benefit is not limited to multi-channel inputs and is complementary to cross-channel modeling.

Replacing PSD-MoE with the learned auxiliary-loss-based router consistently yields higher errors. Under the same training configuration, the deterministic spectral router also introduces no learned routing parameters and reduces the average pre-training time from 2.45 to 2.17 seconds per step. Finally, we move the CD layer from the default top block to progressively shallower positions, where CD Layer@ k th denotes placement at the k -th of the six backbone layers. Errors increase monotonically on both multi-channel datasets as the

Table 4: Corpus-matched comparison on zero-shot forecasting. SOTER and the two corpus-matched MIRA-Large variants are trained for one full epoch on the same physiological corpus under the same training budget. Results are reported as the mean and standard deviation over four input/output window pairs and entries are rescaled by 10^2 , as in Table 1. Lower values are better.

Method	MIT-BIH		ScientiSST MOVE	
	RMSE	MAE	RMSE	MAE
SOTER	1.29± 0.22	0.62± 0.10	2.26± 0.12	1.17± 0.04
MIRA-Large	2.20 ± 0.14	1.54 ± 0.12	4.76 ± 0.16	3.73 ± 0.12
MIRA-Large (scratch)	2.03 ± 0.17	1.38 ± 0.11	3.85 ± 0.20	2.91 ± 0.18
MIRA-Large (continued)	1.79 ± 0.19	1.08 ± 0.17	3.12 ± 0.19	2.37 ± 0.14

CD layer is moved downward, whereas HeartRate remains nearly unchanged. This trend suggests that, within SOTER, cross-channel interaction is most effective after the channel-specific temporal representations have been sufficiently refined.

4.5 Corpus-Matched Comparison

The remaining question is whether the pre-training corpus alone explains this advantage. We therefore pre-train MIRA-Large on the same physiological corpus under the same one-epoch training budget and optimization schedule, both from scratch and by continued pre-training. Here, MIRA-Large denotes the officially released checkpoint, MIRA-Large (scratch) is randomly initialized and trained on our corpus, and MIRA-Large (continued) is initialized from the official checkpoint and further pre-trained on the same corpus. All models are evaluated using the same zero-shot forecasting protocol and input/output window settings. The results are reported in Table 4. Training MIRA-Large on the physiological corpus improves upon its official checkpoint, with continued pre-training producing the strongest MIRA variant on both datasets. Nevertheless, SOTER remains ahead on all four metrics, including reductions over MIRA-Large (continued) from 1.79 to 1.29 RMSE on MIT-BIH and from 3.12 to 2.26 on ScientiSST MOVE. These results indicate that corpus matching contributes to performance, but the corpus alone does not account for SOTER’s gains on the evaluated datasets.

4.6 Interpretability of PSD-Guided Routing

To examine whether the spectral statistic used by the router reflects the frequency content of the underlying signal, we analyze three signals from distinct regimes: high-frequency ECG from MIT-BIH (S1), mid-frequency PPG from IEEE PPG (S2), and low-frequency heart rate from HeartRate (S3). We sample five segments from each signal, pass them through SOTER, and record the gate-assignment vector at the topmost PSD-guided MoE layer. Because the router operates on latent representations rather than raw inputs, the spectral structure available after the attention layers need not directly preserve the input frequency characteristics.

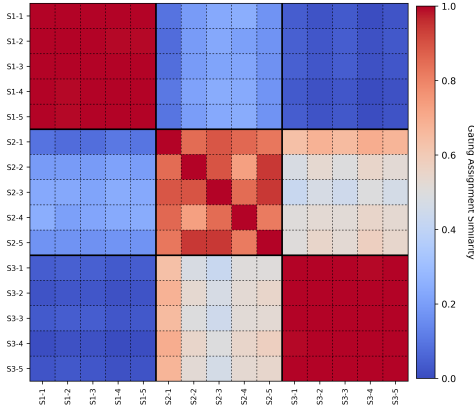


Figure 3: Pairwise cosine similarity between gate-assignment vectors for three signal types (S1-S3), with five segments sampled from each signal using seed 42. The block-diagonal structure indicates that signals from different frequency regimes receive separable routing assignments.

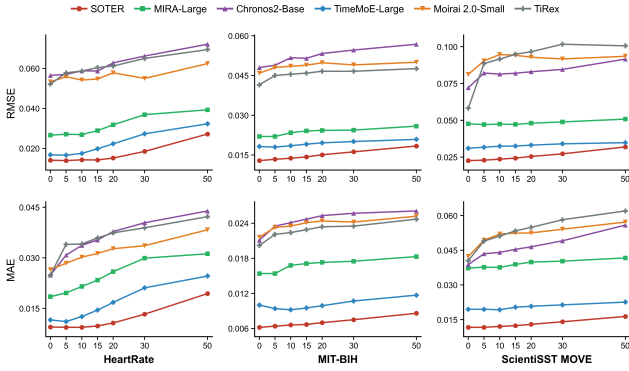


Figure 4: Zero-shot forecasting under additive Gaussian noise. Noise with standard deviation $\beta\sigma_{\text{signal}}$ is injected into the input context, where σ_{signal} is the per-channel signal scale estimated from the training split and $\beta \in \{0.05, \dots, 0.5\}$. RMSE and MAE are computed against the clean future on three datasets.

Figure 3 nevertheless exhibits clear within-signal similarity and cross-signal separation. Segments from the same signal form high-similarity blocks, S1 and S3 are well separated, and S2 lies between these two regimes. This pattern indicates that the latent spectral statistic remains associated with the spectral characteristics of the underlying signals, providing an explicit and inspectable basis for expert routing.

4.7 Noise Robustness Analysis

To emulate acquisition noise in real-world wearable sensing, we inject scale-proportional additive Gaussian noise into the observed context and evaluate zero-shot prediction against the clean future (Figure 4). Across all six noise levels, both metrics, and all three datasets, SOTER achieves the lowest prediction error among the

Table 5: OOD forecasting performance at different levels of cumulative pre-training exposure. Results are the mean and standard deviation over four input/output windows and are rescaled by 10^2 . Lower values are better.

Exposure	MIT-BIH		SciencisSST MOVE		WDD	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
10%	1.80 $\pm$ 0.20	1.35 $\pm$ 0.34	3.33 $\pm$ 0.30	2.21 $\pm$ 0.31	4.89 $\pm$ 0.16	3.41 $\pm$ 0.26
25%	1.52 $\pm$ 0.15	1.05 $\pm$ 0.13	2.59 $\pm$ 0.10	1.80 $\pm$ 0.12	3.16 $\pm$ 0.25	2.19 $\pm$ 0.41
50%	1.32 $\pm$ 0.23	0.88 $\pm$ 0.07	2.44 $\pm$ 0.14	1.39 $\pm$ 0.15	2.69 $\pm$ 0.19	1.47 $\pm$ 0.13
75%	1.29 $\pm$ 0.25	0.63 $\pm$ 0.09	2.32 $\pm$ 0.15	1.25 $\pm$ 0.08	2.22 $\pm$ 0.19	1.17 $\pm$ 0.12
100%	1.29 $\pm$ 0.22	0.62 $\pm$ 0.10	2.26 $\pm$ 0.12	1.17 $\pm$ 0.04	2.19 $\pm$ 0.19	1.05 $\pm$ 0.05

evaluated methods. Even under the strongest corruption level, $\beta = 0.5$, its performance remains comparable to or better than that of the baselines evaluated on clean inputs ($\beta = 0$), demonstrating strong robustness to additive acquisition noise.

4.8 Cumulative Pre-training Exposure Analysis

To examine how OOD transfer evolves over the single-epoch pre-training run, we evaluate checkpoints corresponding to cumulative corpus exposures of 10%, 25%, 50%, 75%, and 100% under the same four forecasting-window settings, as shown in Table 5. Across all three OOD datasets, increasing cumulative pre-training exposure consistently reduces both RMSE and MAE, indicating progressively improved downstream transfer over the course of training. The rate of saturation, however, is dataset- and metric-dependent. On MIT-BIH, the RMSE at 50% exposure is already within 2.3% of the final result, whereas the MAE approaches its final value only after 75% exposure. On SciencisSST MOVE, the RMSE and MAE at 50% exposure remain 8.0% and 18.8% above their final values, respectively; these gaps narrow to 2.7% and 6.8% at 75%. WDD exhibits the strongest dependence on later-stage pre-training, particularly in MAE. Overall, OOD transfer improves consistently as pre-training progresses, while the marginal performance gains gradually diminish at dataset- and metric-dependent rates.

5 Conclusion

We presented SOTER, a generative time-series foundation model that jointly models cross-channel coupling, spectrum-guided expert specialization, and continuous-time dynamics for wearable physiological signals. Across out-of-distribution forecasting, linear-probe classification, and causal continuous-time imputation, a single pre-trained backbone achieves strong and consistent transfer while activating only 20.29M parameters per inference step, and remains robust under substantial missingness and additive acquisition noise. Component ablations support the non-redundant roles of the three modeling mechanisms, while the full-budget corpus-matched comparison shows that physiological pre-training improves MIRA-Large but does not alone account for SOTER’s advantage. These findings suggest that aligning model structure with the channel organization, spectral heterogeneity, and continuous-time nature of physiological processes is an effective direction for domain-specialized time-series foundation models. Future work will extend this framework to broader physiological modalities and populations, larger-scale pre-training, and prospective clinical evaluation.

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A Datasets and Statistics

We briefly describe the datasets used in this work. SOTER is pre-trained on a wearable physiological corpus containing approximately 226 billion time points, assembled from five publicly available and de-identified datasets: MIMIC-III Waveform [22, 39], Sleep-EDF [23], PTB-XL [55, 56], WESAD [52], and Chapman-ECG [63, 64]. Evaluation is conducted on six publicly available physiological datasets recorded using wearable or ambulatory sensing devices: HeartRate [54], MIT-BIH [40], ScientisST MOVE [3], IEEE PPG [62], MMASH [46, 47], and WDD [18, 19]. We introduce the pre-training corpora first, followed by the evaluation datasets.

A.1 Pre-training Corpora

MIMIC-III Waveform. The MIMIC-III Waveform Database comprises 67,830 record sets from approximately 30,000 intensive-care-unit patients. Most record sets pair digitized physiological waveforms—typically ECG, arterial blood pressure, respiration, and PPG—with periodically sampled numerical vital signs, forming quasi-continuous patient recordings spanning from days to several weeks. A subset is matched and time-aligned with the MIMIC-III Clinical Database.

Sleep-EDF. Sleep-EDF contains 197 whole-night polysomnographic recordings comprising EEG, EOG, chin EMG, and event markers, with some recordings additionally providing respiration and body temperature. Each recording is accompanied by an expert-scored hypnogram annotated according to the Rechtschaffen and Kales standard.

PTB-XL. PTB-XL is a large clinical electrocardiography dataset containing 21,799 ten-second, 12-lead ECG recordings from 18,869 patients. Each record was annotated by up to two cardiologists with potentially multiple statements drawn from 71 SCP-ECG-conformant diagnostic, form, and rhythm categories, together with demographic and signal-property metadata.

WESAD. WESAD is a multimodal wearable dataset for stress and affect detection, recorded from wrist- and chest-worn devices on 15 subjects. It includes blood volume pulse, ECG, electrodermal activity, electromyography, respiration, body temperature, and three-axis acceleration, and covers three affective states: neutral, stress, and amusement.

Chapman-ECG. The Chapman-ECG database provides 12-lead ECG recordings from 45,152 patients sampled at 500 Hz. It covers multiple common cardiac rhythms and additional cardiovascular conditions, with labels provided by professional experts, and supports the development and evaluation of methods for arrhythmia and related cardiovascular analysis.

A.2 Evaluation Datasets

HeartRate. Drawn from the Monash Time Series Extrinsic Regression Archive, HeartRate pairs short photoplethysmography (PPG) segments with continuous heart-rate targets. It serves as the single-channel physiological benchmark in our evaluation.

MIT-BIH. The MIT-BIH Arrhythmia Database contains 48 half-hour, two-channel ambulatory ECG excerpts from 47 subjects, digitized at 360 Hz per channel with 11-bit resolution over a 10 mV

Table 6: Type, number of feature channels, and normalized permutation entropy (NPE) of the six evaluation datasets. SC and MC denote single-channel and multi-channel datasets, respectively. Higher NPE indicates less predictable temporal dynamics.

Dataset	Type	#Channels	NPE
HeartRate	SC	1	0.469
MIT-BIH	MC	2	0.878
ScientiSST MOVE	MC	5	0.643
IEEE PPG	MC	5	0.646
MMASH	MC	6	0.334
WDD	MC	7	0.618

range. Each beat was independently annotated by two or more cardiologists, yielding approximately 110,000 reference annotations.

ScientiSST MOVE. ScientiSST MOVE records everyday activities—including lifting a chair, greeting, gesticulating, walking, and running—from 17 healthy volunteers using a chestband, an armband, and an Empatica E4 wristband. The dataset includes multi-channel EDA, PPG, and ECG signals, together with bicep EMG, wrist temperature, and actigraphy, providing physiological measurements under dynamic and less-controlled conditions.

IEEE PPG. Originating from the IEEE Signal Processing Cup 2015, IEEE PPG targets heart-rate estimation during physical exercise. It contains 3,096 multi-channel sequences comprising two-channel wrist PPG, three-axis acceleration, and one-channel chest ECG, recorded simultaneously at 125 Hz from subjects aged 18–35.

MMASH. MMASH provides 24-hour recordings of continuous beat-to-beat heart data, triaxial acceleration, sleep quality, physical activity, and psychological characteristics from 22 healthy participants, together with saliva biomarkers and activity logs. It supports analysis of the relationships among physical activity, sleep, and physiological state.

WDD. WDD records physiological signals during structured acute-stress induction and aerobic or anaerobic exercise using the Empatica E4 wearable device. It includes blood volume pulse, motion-based activity, skin temperature, and electrodermal activity. The dataset comprises 36 volunteers for stress sessions, 30 for aerobic exercise, and 31 for anaerobic exercise.

Table 6 summarizes the channel configuration and normalized permutation entropy (NPE) of each evaluation dataset. We use NPE as a scalar indicator of temporal complexity, with larger values corresponding to less predictable dynamics. MMASH has the lowest NPE among the six datasets (0.334), whereas MIT-BIH has the highest (0.878), providing additional context for the forecasting results discussed in the main paper. All datasets used in this study are **publicly available and de-identified**, and no personally identifiable information was accessed.

B Pre-training Setup and Logs

This section details the pre-training configuration of SOTER in sufficient depth to support reproduction and visualizes its convergence.

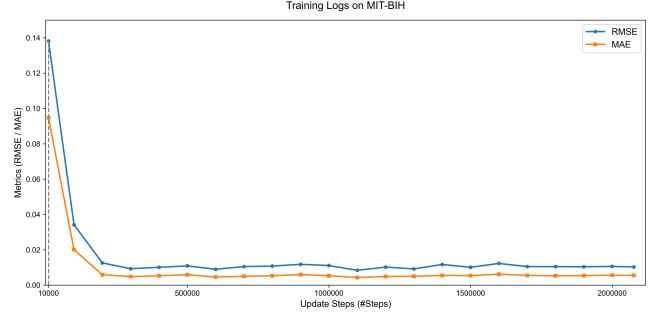


Figure 5: Pre-training convergence on the MIT-BIH out-of-distribution quick evaluation. Every 100,000 optimization steps, RMSE and MAE in the normalized space are computed on 32 held-out MIT-BIH samples using input/output windows of 128/64. Both metrics decrease sharply during the early evaluation intervals and subsequently remain on a low and stable plateau throughout the remainder of the single-epoch run.

Compute and data pipeline. SOTER is pre-trained for a single epoch on the full corpus of approximately 226 billion time points, amounting to 2,076,572 optimization steps, on 8× NVIDIA H200 GPUs. We use a per-GPU batch size of 8 with no gradient accumulation, resulting in a global batch size of 64. The maximum sequence length is 512, and data loading uses 12 workers. Training is conducted in bf16 mixed precision, and attention is computed using an eager, non-fused implementation. The windowed-sample cache is pre-warmed before optimization begins.

Optimization. We optimize SOTER using AdamW with a learning rate of 3×10^{-4} , $\beta = (0.9, 0.95)$, $\epsilon = 10^{-8}$, and weight decay of 0.01. We employ a cosine-annealing schedule with linear warmup. The warmup is deliberately short ($warmup_steps = 20$), because the single-epoch run contains a large number of optimization steps, and the cosine schedule spans the actual number of steps in one epoch. No gradient clipping is applied, and all runs use seed 42.

Decoder solver and PSD routing. The continuous-time decoder is integrated using the Dormand–Prince 5(4) solver (dopri5) with a tolerance of 10^{-5} . The PSD-guided router is implemented using a parallel, strictly causal prefix discrete Fourier transform (DFT). The effects of the DFT implementation, solver family, and solver tolerance are examined in Appendix C.

Convergence monitoring. Every 100,000 optimization steps, we conduct a lightweight out-of-distribution evaluation on MIT-BIH. We evaluate 32 held-out samples using input and output window lengths of 128 and 64, respectively, with seed 123. RMSE and MAE are reported in the normalized space, where the per-channel Min-Max normalization parameters are estimated exclusively from the MIT-BIH training split. Figure 5 visualizes the resulting trajectory.

As shown in Figure 5, both RMSE and MAE decrease rapidly during the earliest evaluation intervals and subsequently stabilize at low values for the remainder of the single-epoch run, indicating stable pre-training convergence.

Table 7: Sensitivity analysis of the DFT implementation, continuous-time solver, and solver tolerance. Each configuration is evaluated in a separate 100,000-step sensitivity run, independent of full pre-training, on zero-shot forecasting over MIT-BIH using 256 samples, input/output windows of 128/64, and seed 42. RMSE and MAE are reported in the normalized space, and latency denotes the average optimization time per step within this sensitivity run. Bold and underlined values indicate the best and second-best results in each metric, respectively. The highlighted row is the configuration adopted by SOTER.

DFT Implementation	Solver	Tolerance	RMSE ↓	MAE ↓	Latency (s/step) ↓
For Loop	dopri5	10^{-6}	0.0477	0.0350	9.88
For Loop	dopri5	10^{-5}	0.0470	0.0345	9.92
For Loop	dopri5	10^{-4}	0.0487	0.0359	9.90
For Loop	dopri5	10^{-3}	0.0504	0.0373	9.80
For Loop	rk4	10^{-6}	0.0482	0.0353	9.79
For Loop	rk4	10^{-5}	0.0484	0.0355	9.04
For Loop	rk4	10^{-4}	0.0487	0.0358	9.05
For Loop	rk4	10^{-3}	0.0488	0.0359	9.08
Parallel Prefix	dopri5	10^{-6}	0.0483	0.0353	2.18
Parallel Prefix	dopri5	10^{-5}	0.0473	0.0347	2.17
Parallel Prefix	dopri5	10^{-4}	0.0478	0.0350	2.09
Parallel Prefix	dopri5	10^{-3}	0.0488	0.0359	<u>2.08</u>
Parallel Prefix	rk4	10^{-6}	0.0479	0.0352	2.14
Parallel Prefix	rk4	10^{-5}	0.0481	0.0355	2.12
Parallel Prefix	rk4	10^{-4}	0.0482	0.0356	2.17
Parallel Prefix	rk4	10^{-3}	0.0479	0.0351	2.03

C Key Sensitivity Analysis

To balance predictive performance and training efficiency, we analyze the effects of the discrete Fourier transform (DFT) implementation, continuous-time solver, and solver tolerance. We compare a parallel-prefix implementation with a loop-based implementation for the causal prefix DFT, consider dopri5 and rk4 as the numerical solvers, and evaluate tolerances of 10^{-6} , 10^{-5} , 10^{-4} , and 10^{-3} .

All configurations are evaluated after 100,000 optimization steps using zero-shot forecasting on MIT-BIH with 256 samples, input/output windows of 128/64, and random seed 42. We report RMSE and MAE in the normalized space, together with the average latency per optimization step. Table 7 summarizes the results.

The loop-based DFT consistently incurs substantially higher latency than the parallel-prefix implementation. Although the lowest forecasting errors are obtained by the loop-based DFT with dopri5 and a tolerance of 10^{-5} , this configuration requires approximately four times the per-step latency of its parallel-prefix counterpart. The fastest configuration combines the parallel-prefix DFT, rk4, and a tolerance of 10^{-3} , but yields slightly higher prediction errors. We therefore adopt the parallel-prefix DFT with dopri5 and a tolerance of 10^{-5} , which adds only 0.14 seconds per step relative to the fastest configuration while providing a more favorable accuracy-efficiency trade-off. This choice is consistent with the pre-training configuration described in Appendix B.

D From Neural CDE to Neural ODE

This section formalizes the relationship between the two operating regimes of SOTER’s terminal continuous-time decoder. The Neural ODE used for future forecasting is not implemented as a separate module; rather, it is obtained from the observation-guided

Neural CDE regime by setting the external control path to zero. Both regimes share the same vector field and parameter set θ . The resulting reduction is an algebraic identity rather than a numerical approximation.

The notation follows the main paper. The terminal backbone state at the last observed timestamp t_T is denoted by h_T , and t_{T+1} denotes the queried timestamp. The formulation below provides the implementation-level realization of Eqs. (7) and (8).

D.1 Setup and Notation

The autoregressive backbone produces the terminal hidden state

$$h_T = h(t_T) \in \mathbb{R}^H, \quad (9)$$

where H denotes the hidden dimension. The continuous-time decoder propagates h_T toward the queried timestamp t_{T+1} .

We define the integration interval as

$$\begin{aligned} \delta t &= t_{T+1} - t_T, \\ \Delta t &= \max(\delta t, \epsilon), \quad \epsilon = 10^{-5}, \end{aligned} \quad (10)$$

where the lower bound ϵ is a numerical safeguard against a degenerate zero-length interval. For all non-degenerate queries, $\Delta t = \delta t$.

Normalized integration variable. Numerical integration is performed over a normalized variable $s \in [0, 1]$. Physical time is related to s through the affine reparameterization

$$\begin{aligned} t(s) &= t_T + s\Delta t, \quad s \in [0, 1], \\ \frac{dt}{ds} &= \Delta t. \end{aligned} \quad (11)$$

Consequently, $s = 0$ corresponds to the last observed timestamp t_T , and $s = 1$ corresponds to the queried endpoint.

Vector field. The decoder dynamics are generated by a single multilayer perceptron f_θ with parameters θ :

$$\begin{aligned} f_\theta: [0, 1] \times \mathbb{R}^H \times \mathbb{R}^{d_z} &\rightarrow \mathbb{R}^H, \\ f_\theta(s, h, z) &= \text{MLP}_\theta([s; h; z]). \end{aligned} \quad (12)$$

Here, s is the normalized time coordinate, $h \in \mathbb{R}^H$ is the latent state, and $z \in \mathbb{R}^{d_z}$ is an external control input. In our configuration, $d_z = 1$ for each independently processed channel. The parameter set θ is shared across all samples and across both decoder regimes.

Control path. The control is a time-dependent function

$$z: [t_T, t_T + \Delta t] \rightarrow \mathbb{R}^{d_z}. \quad (13)$$

At normalized integration coordinate s , the decoder evaluates the control at the corresponding physical timestamp $t(s) = t_T + s\Delta t$. The normalized time feature supplied to f_θ and the physical-time argument supplied to $z(\cdot)$ are therefore kept distinct.

D.2 Unified Terminal Dynamics

Both decoder regimes are instances of the same initial-value problem. Using the normalized variable s and the Jacobian $dt/ds = \Delta t$, the decoder solves

$$\begin{aligned} \frac{dh}{ds}(s) &= \Delta t f_\theta(s, h(s), z(t_T + s\Delta t)), \\ h(0) &= h_T. \end{aligned} \quad (14)$$

The endpoint state is

$$h(1) = h_T + \int_0^1 \Delta t f_\theta(s, h(s), z(t_T + s\Delta t)) ds. \quad (15)$$

For clarity, define the corresponding physical-time state

$$\bar{h}(t) := h\left(\frac{t - t_T}{\Delta t}\right). \quad (16)$$

Undoing the reparameterization in Eq. (11) gives

$$\bar{h}(t_{T+1}) = h_T + \int_{t_T}^{t_{T+1}} f_\theta\left(\frac{t - t_T}{\Delta t}, \bar{h}(t), z(t)\right) dt, \quad (17)$$

for non-degenerate intervals. Equations (15) and (17) describe the same trajectory in normalized and physical-time coordinates, respectively. The normalized form in Eq. (14) is the one solved numerically.

D.3 The Two Decoder Regimes

The forecasting and imputation regimes differ only in the control path used in Eq. (14).

Definition D.1 (Observation-guided regime). For causal imputation, the available observations preceding the query are interpolated using a natural cubic spline $S(\cdot)$, and the control is defined as

$$z(t) = S(t). \quad (18)$$

The latent dynamics are therefore conditioned on the observable causal prefix through the external control input.

Definition D.2 (Empty-control forecasting regime). For future forecasting, no observation-derived control is available beyond the terminal context timestamp. The implementation represents this absence of control using

$$z(t) \equiv \mathbf{0} \in \mathbb{R}^{d_z}. \quad (19)$$

The zero vector in Eq. (19) is a valid input to the same vector field in Eq. (12). Therefore, switching from observation-guided imputation to forecasting does not alter the decoder architecture or introduce a second parameter set.

D.4 Exact Reduction

We next show that the empty-control regime reduces exactly to a time-augmented Neural ODE.

Input layer under zero control. Let the first affine layer of MLP_θ have weight $W^{(1)}$ and bias $b^{(1)}$. Partition its columns according to the three input blocks $[s; h; z]$:

$$W^{(1)} = \begin{bmatrix} W_s^{(1)} & W_h^{(1)} & W_z^{(1)} \end{bmatrix}. \quad (20)$$

The corresponding pre-activation is

$$a^{(1)}(s, h, z) = W_s^{(1)}s + W_h^{(1)}h + W_z^{(1)}z + b^{(1)}. \quad (21)$$

When $z = \mathbf{0}$,

$$a^{(1)}(s, h, \mathbf{0}) = W_s^{(1)}s + W_h^{(1)}h + b^{(1)}, \quad (22)$$

because $W_z^{(1)}\mathbf{0} = \mathbf{0}$. Hence, under zero control, the first-layer output and all subsequent activations depend only on (s, h) and the shared parameters θ .

Reduced vector field. Define

$$\tilde{f}_\theta(s, h) := f_\theta(s, h, \mathbf{0}). \quad (23)$$

Substituting Eq. (19) into Eq. (14) yields

$$\begin{aligned} \frac{dh}{ds}(s) &= \Delta t \tilde{f}_\theta(s, h(s)), \\ h(0) &= h_T. \end{aligned} \quad (24)$$

Its endpoint solution is

$$h(1) = h_T + \int_0^1 \Delta t \tilde{f}_\theta(s, h(s)) ds. \quad (25)$$

Equation (24) is a time-augmented Neural ODE: its vector field depends only on the normalized time coordinate and the latent state. It uses the same parameter set θ as Eq. (14); the control-related column block $W_z^{(1)}$ remains part of the shared network but contributes zero under Eq. (19).

The reduction follows directly and pointwise:

$$\begin{aligned} f_\theta(s, h(s), z(t_T + s\Delta t)) &= f_\theta(s, h(s), \mathbf{0}) \\ &= \tilde{f}_\theta(s, h(s)). \end{aligned} \quad (26)$$

No parameter is added, removed, or updated when switching between the two regimes. Therefore, the reduction is an exact algebraic identity, not an approximation induced by the numerical solver.

Well-posedness. Suppose that f_θ is continuous in (s, z) and uniformly Lipschitz continuous in h over the compact integration interval. The natural cubic spline control is continuous, while the empty control is constant. Under these conditions, the Picard–Lindelöf theorem ensures that both Eq. (14) and its reduction in Eq. (24) admit unique solutions. In particular, fixing $z \equiv \mathbf{0}$ preserves the Lipschitz property in h and requires no additional well-posedness assumptions. An MLP with finite weights and Lipschitz activations satisfies the required Lipschitz condition.

Globally disabled control channel. If the decoder is instantiated with $d_z = 0$, the control block is omitted from the concatenated input, and its vector field takes the form

$$f_\theta(s, h) = \text{MLP}_\theta([s; h]). \quad (27)$$

Functionally, this is the same reduced form as Eq. (23), with the zero-contributing control columns omitted at construction time. Thus, both a runtime empty control path and a globally disabled control channel result in time-augmented Neural ODE dynamics.

Terminology. The formulation in Eq. (12) injects the control value $z(t)$ as an input to the vector field:

$$\frac{dh}{dt} = f_\theta(\cdot, h(t), z(t)). \quad (28)$$

This is a control-as-input neural differential equation. It is related to, but mathematically distinct from, the commonly used bilinear Neural CDE formulation

$$dh(t) = g_\theta(h(t)) dX(t), \quad (29)$$

in which the vector field acts on increments of the control path. When X is differentiable, the latter can be written as

$$\frac{dh}{dt} = g_\theta(h(t)) \dot{X}(t). \quad (30)$$

SOTER instead supplies the control value jointly with time and state to a single MLP. Throughout this work, Neural CDE-style dynamics refer to this control-as-input controlled neural differential equation. The zero-control reduction established above applies specifically to this formulation.

Task interpretation. The reduction is consistent with the semantics of the two downstream tasks. In future forecasting, no observation exists beyond the context boundary from which to construct a control path; setting $z \equiv \mathbf{0}$ therefore yields autonomous extrapolation from the terminal state h_T . In causal imputation, the observations available before the query define the spline control $z = S(\cdot)$, which guides the same latent dynamics toward the missing timestamp.

Numerical solver. Both regimes are integrated over $s \in [0, 1]$ using the same adaptive Dormand–Prince 5(4) solver (dopri5), with absolute and relative tolerances of 10^{-5} . The selection of the solver and tolerance is detailed in Appendix C. The algebraic reduction is independent of the numerical solver and its tolerance; these choices affect only the numerical approximation of Eqs. (15) and (25).

D.5 Summary

The forecasting decoder in Eq. (24) is obtained by evaluating the observation-guided decoder in Eq. (14) at the zero control path. The two regimes share the same network structure and parameter set θ . Consequently, a single frozen continuous-time decoder supports both observation-guided causal imputation and autonomous future forecasting without switching models or introducing task-specific decoder parameters.

E System Overhead Analysis

To characterize the practical inference overhead, we measure the average single-sample latency of representative models under the

same hardware and software environment and report their parameter counts using the same convention as in the main paper. TimeMoE-Large and TimeMoE-Base require 3.72 s and 3.92 s per sample, with 453.2M and 113.3M parameters, respectively. MIRA-Large contains 339.9M parameters and incurs the highest latency of 8.97 s, whereas Aurora achieves the lowest latency of 0.138 s with 260.8M parameters. SOTER requires 6.09 s per sample while activating only 20.29M parameters. It is therefore slower than Aurora and the two TimeMoE variants, but faster than MIRA-Large and substantially more compact than all compared models. These results also show that wall-clock latency is not determined by parameter count alone: SOTER’s adaptive continuous-time integration introduces non-negligible numerical overhead, while architectural structure and implementation characteristics substantially influence runtime. The comparison should thus be interpreted as a practical accuracy–compactness–latency trade-off rather than a universal ranking of inference efficiency.

F Complete Per-Window Zero-Shot Forecasting Results

This section reports the per-window zero-shot forecasting measurements underlying Table 1. Unlike the main table, which reports the mean and sample standard deviation across the supported input/output windows after rescaling all error values by 10^2 for readability, Tables 8–11 report the unscaled RMSE and MAE values in the normalized space for each individual window. The aggregate statistics in Table 1 are computed from the underlying full-precision measurements before rounding; consequently, recomputing them from the four-decimal values reported below may produce a difference in the final displayed digit. Timer supports only the two longest input/output windows, 96/48 and 128/64.

G Continuous-Time Imputation across Missing Rates

Figure 2 in the main paper visualizes the mean continuous-time imputation errors over three repeated runs at missing rates of 25%, 50%, and 75%. For visual clarity, the figure plots only the mean value of each model and configuration. In contrast, Tables 12–14 report the complete numerical results, including the corresponding standard deviations and the full set of evaluated baselines. All entries are reported as the mean and standard deviation over three runs with different random seeds in the normalized space.

At missing rates of 25% and 50%, SOTER achieves the best result on 10 of the 12 dataset–metric combinations and ranks second on the remaining two, both on MMASH, where TiRex performs best. At a missing rate of 75%, SOTER achieves the lowest error on all 12 combinations. Several baselines also exhibit substantially greater degradation as the missing rate increases. For example, on MIT-BIH, the RMSE of Chronos2-Base increases from 0.0578 at 25% missingness to 0.4850 at 75%, while that of Sundial-Base increases from 0.1063 to 0.4105. Over the same range, SOTER increases from 0.0183 to 0.1502. These results are consistent with the ability of SOTER’s continuous-time decoder to query a target timestamp directly from the available causal prefix.

Table 8: Per-window zero-shot forecasting results for the 48/24 input/output setting. Unscaled RMSE and MAE values are reported in the normalized space. Timer does not support this setting and is marked “/”. Bold and underlined values denote the best and second-best results in each column, respectively.

Method	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)		IEEE PPG (MC)		MMASH (MC)		WDD (MC)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Timer	/	/	/	/	/	/	/	/	/	/	/	/
TTM	0.0397	0.0214	0.0398	0.0192	0.0948	0.0715	0.0934	0.0629	0.0106	0.0079	0.0207	0.0167
Lag-Llama	0.1967	0.1554	0.4906	0.4876	0.5134	0.4726	0.5187	0.5082	0.1849	0.1769	0.4348	0.4145
TiRex	0.0367	0.0171	0.0448	0.0208	0.0634	0.0467	0.0919	0.0536	0.0010	<u>0.0003</u>	0.0106	0.0078
MOMENT-Large	0.0620	0.0366	0.0419	0.0220	0.1147	0.0927	0.1022	0.0618	0.0036	<u>0.0015</u>	0.0114	<u>0.0060</u>
MOMENT-Base	0.0685	0.0429	0.0470	0.0247	0.1157	0.0920	0.1024	0.0621	0.0032	0.0016	0.0113	0.0063
MOMENT-Small	0.0651	0.0392	0.0426	0.0223	0.1130	0.0902	0.1049	0.0633	0.0044	0.0017	0.0109	0.0058
MoiraiMoE-Base	0.0583	0.0246	0.0512	0.0228	0.1397	0.0963	0.1251	0.0696	0.0022	0.0009	0.0114	0.0101
MoiraiMoE-Small	0.0619	0.0280	0.0509	0.0227	0.1430	0.0992	0.1296	0.0713	0.0026	0.0010	0.0112	0.0097
Moirai2.0-Small	0.0527	0.0214	0.0479	0.0217	0.0877	0.0432	0.0947	0.0548	<u>0.0007</u>	<u>0.0003</u>	0.0096	0.0084
TimesFM 1.0	0.0472	0.0231	0.0441	0.0218	0.1165	0.0853	0.0950	0.0538	<u>0.0027</u>	<u>0.0008</u>	0.0090	0.0066
TimesFM 2.0	0.0545	0.0256	0.0464	0.0211	0.1170	0.0832	0.1054	0.0585	0.0010	<u>0.0003</u>	<u>0.0089</u>	0.0065
TimesFM 2.5	0.0521	0.0218	0.0429	0.0201	0.1182	0.0833	0.0992	0.0559	0.0003	0.0001	0.0093	0.0072
TimeMoE-Large	0.0266	0.0163	<u>0.0197</u>	<u>0.0099</u>	<u>0.0319</u>	<u>0.0203</u>	<u>0.0173</u>	0.0108	0.0339	0.0288	0.0167	0.0125
TimeMoE-Base	<u>0.0234</u>	<u>0.0162</u>	<u>0.0206</u>	<u>0.0123</u>	<u>0.0332</u>	<u>0.0213</u>	0.0172	<u>0.0112</u>	0.0358	0.0287	0.0204	0.0162
Chronos2-Base	0.0523	0.0236	0.0632	0.0238	0.0801	0.0410	0.1090	0.0616	0.0012	0.0005	0.0099	0.0084
Chronos2-Small	0.0549	0.0252	0.0519	0.0224	0.1156	0.0798	0.1034	0.0597	0.0015	0.0006	0.0098	0.0083
Sundial-Base	0.0678	0.0313	0.0409	0.0198	0.1122	0.0822	0.0907	0.0525	0.0156	0.0124	0.0151	0.0109
Aurora	0.0853	0.0456	0.0509	0.0274	0.1604	0.1262	0.1083	0.0671	0.0197	0.0085	0.0107	0.0062
MIRA-Large	0.0478	0.0343	0.0234	0.0167	0.0494	0.0387	0.0266	0.0204	0.0653	0.0581	0.0334	0.0295
SOTER	0.0150	0.0106	0.0157	0.0075	0.0244	0.0121	0.0246	<u>0.0112</u>	0.0163	0.0104	0.0086	0.0072

Table 9: Per-window zero-shot forecasting results for the 72/36 input/output setting. Unscaled RMSE and MAE values are reported in the normalized space. Timer does not support this setting and is marked “/”. Bold and underlined values denote the best and second-best results in each column, respectively.

Method	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)		IEEE PPG (MC)		MMASH (MC)		WDD (MC)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Timer	/	/	/	/	/	/	/	/	/	/	/	/
TTM	0.0485	0.0271	0.0418	0.0205	0.0959	0.0739	0.0929	0.0624	0.0120	0.0074	0.0175	0.0124
Lag-Llama	0.1989	0.1567	0.4886	0.4857	0.5063	0.4649	0.5165	0.5059	0.1837	0.1760	0.4385	0.4146
TiRex	0.0526	0.0272	0.0407	0.0203	0.0612	0.0426	0.0927	0.0567	0.0025	0.0009	0.0099	0.0084
MOMENT-Large	0.0589	0.0311	0.0440	0.0226	0.0894	0.0734	0.0978	0.0618	0.0052	0.0024	0.0107	<u>0.0060</u>
MOMENT-Base	0.0685	0.0361	0.0521	0.0258	0.0930	0.0729	0.1018	0.0638	0.0048	0.0022	0.0108	0.0063
MOMENT-Small	0.0624	0.0311	0.0449	0.0227	0.0881	0.0708	0.1036	0.0649	0.0049	0.0022	0.0108	0.0062
MoiraiMoE-Base	0.0712	0.0342	0.0622	0.0279	0.1352	0.0971	0.1207	0.0734	0.0054	0.0022	0.0125	0.0107
MoiraiMoE-Small	0.0725	0.0345	0.0563	0.0256	0.1349	0.0967	0.1213	0.0728	0.0060	0.0025	0.0118	0.0099
Moirai2.0-Small	0.0506	0.0307	0.0506	0.0234	0.0789	0.0435	0.1021	0.0617	0.0070	0.0026	0.0098	0.0085
TimesFM 1.0	0.0708	0.0317	0.0426	0.0207	0.1154	0.0832	0.0820	0.0527	0.0035	0.0010	0.0095	0.0068
TimesFM 2.0	0.0708	0.0331	0.0469	0.0215	0.1154	0.0836	0.0908	0.0567	0.0023	<u>0.0007</u>	0.0094	0.0068
TimesFM 2.5	0.0563	0.0260	0.0418	0.0201	0.1161	0.0849	0.0857	0.0535	0.0007	0.0003	<u>0.0092</u>	0.0074
TimeMoE-Large	<u>0.0154</u>	<u>0.0108</u>	<u>0.0179</u>	<u>0.0096</u>	<u>0.0315</u>	<u>0.0199</u>	0.0187	<u>0.0109</u>	0.0305	0.0262	<u>0.0157</u>	0.0121
TimeMoE-Base	0.0176	0.0125	0.0197	0.0124	0.0353	0.0225	<u>0.0188</u>	0.0113	0.0310	0.0255	0.0206	0.0159
Chronos2-Base	0.0464	0.0208	0.0495	0.0234	0.0747	0.0402	<u>0.1018</u>	0.0635	<u>0.0012</u>	<u>0.0007</u>	0.0109	0.0062
Chronos2-Small	0.0611	0.0272	0.0507	0.0244	0.1175	0.0813	0.1041	0.0647	<u>0.0019</u>	<u>0.0008</u>	0.0108	0.0062
Sundial-Base	0.0637	0.0296	0.0423	0.0201	0.1061	0.0793	0.0841	0.0525	0.0119	0.0107	0.0149	0.0102
Aurora	0.0622	0.0362	0.0541	0.0254	0.1282	0.0991	0.0978	0.0625	0.0182	0.0074	0.0100	0.0056
MIRA-Large	0.0285	0.0181	0.0228	0.0158	0.0483	0.0376	0.0253	0.0202	0.0628	0.0559	0.0334	0.0301
SOTER	0.0119	0.0080	0.0136	0.0064	0.0223	0.0119	0.0217	0.0104	0.0127	0.0098	0.0075	0.0062

Table 10: Per-window zero-shot forecasting results for the 96/48 input/output setting. Unscaled RMSE and MAE values are reported in the normalized space. Bold and underlined values denote the best and second-best results in each column, respectively.

Method	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)		IEEE PPG (MC)		MMASH (MC)		WDD (MC)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Timer	0.0801	0.0448	0.0629	0.0365	0.1559	0.1193	0.1254	0.0843	0.0272	0.0209	0.0325	0.0263
TTM	0.0550	0.0309	0.0420	0.0211	0.0966	0.0756	0.0895	0.0582	0.0163	0.0102	0.0216	0.0166
Lag-Llama	0.1985	0.1551	0.4895	0.4865	0.5133	0.4727	0.5176	0.5070	0.1829	0.1754	0.4385	0.4146
TiRex	0.0674	0.0299	0.0432	0.0212	0.0563	0.0370	0.0872	0.0535	0.0048	0.0024	<u>0.0089</u>	0.0082
MOMENT-Large	0.0685	0.0334	0.0418	0.0215	0.0918	0.0757	0.1106	0.0697	0.0082	0.0039	<u>0.0129</u>	0.0067
MOMENT-Base	0.0739	0.0386	0.0478	0.0258	0.0908	0.0718	0.1146	0.0716	0.0083	0.0039	0.0139	0.0075
MOMENT-Small	0.0706	0.0349	0.0441	0.0234	0.0885	0.0704	0.1150	0.0718	0.0082	0.0038	0.0129	0.0066
MoiraiMoE-Base	0.0762	0.0374	0.0554	0.0258	0.1285	0.0922	0.1095	0.0687	0.0101	0.0038	0.0130	0.0103
MoiraiMoE-Small	0.0801	0.0406	0.0548	0.0253	0.1329	0.0963	0.1143	0.0708	0.0092	0.0035	0.0132	0.0109
Moirai2.0-Small	0.0572	0.0283	0.0456	0.0219	0.0798	0.0418	0.0842	0.0532	<u>0.0019</u>	<u>0.0007</u>	0.0112	0.0092
TimesFM 1.0	0.0644	0.0296	0.0446	0.0219	0.1034	0.0771	0.0818	0.0519	<u>0.0032</u>	<u>0.0011</u>	0.0108	0.0071
TimesFM 2.0	0.0710	0.0338	0.0447	0.0216	0.1079	0.0805	0.0868	0.0558	0.0026	0.0010	0.0102	0.0069
TimesFM 2.5	0.0627	0.0278	0.0431	0.0206	0.1071	0.0781	0.0821	0.0517	0.0008	0.0003	0.0110	0.0084
TimeMoE-Large	0.0129	<u>0.0097</u>	<u>0.0183</u>	<u>0.0102</u>	<u>0.0309</u>	<u>0.0193</u>	0.0176	<u>0.0106</u>	0.0295	0.0252	0.0153	0.0119
TimeMoE-Base	0.0170	<u>0.0117</u>	<u>0.0218</u>	<u>0.0144</u>	<u>0.0359</u>	<u>0.0226</u>	<u>0.0180</u>	<u>0.0114</u>	0.0341	0.0266	0.0230	0.0174
Chronos2-Base	0.0702	0.0303	0.0449	0.0209	0.0699	0.0387	0.0852	0.0534	0.0024	0.0009	0.0110	0.0053
Chronos2-Small	0.0546	0.0319	0.0445	0.0212	0.0999	0.0689	0.0854	0.0536	0.0031	0.0012	0.0112	<u>0.0055</u>
Sundial-Base	0.0459	0.0244	0.0430	0.0206	0.1117	0.0847	0.0794	0.0512	0.0101	0.0092	0.0130	0.0091
Aurora	0.0731	0.0398	0.0494	0.0258	0.1424	0.1097	0.1014	0.0650	0.0179	0.0078	0.0121	0.0063
MIRA-Large	0.0164	0.0116	0.0214	0.0139	0.0471	0.0368	0.0252	0.0204	0.0615	0.0537	0.0340	0.0302
SOTER	<u>0.0145</u>	0.0093	0.0114	0.0056	0.0219	0.0117	0.0207	0.0099	0.0112	0.0082	0.0067	<u>0.0055</u>

Table 11: Per-window zero-shot forecasting results for the 128/64 input/output setting. Unscaled RMSE and MAE values are reported in the normalized space. Bold and underlined values denote the best and second-best results in each column, respectively.

Method	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)		IEEE PPG (MC)		MMASH (MC)		WDD (MC)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Timer	0.0979	0.0521	0.0593	0.0363	0.1560	0.1192	0.1330	0.0898	0.0311	0.0228	0.0339	0.0276
TTM	0.0589	0.0332	0.0419	0.0216	0.0967	0.0762	0.0900	0.0589	0.0204	0.0121	0.0232	0.0181
Lag-Llama	0.2041	0.1580	0.4887	0.4857	0.5079	0.4667	0.5173	0.5063	0.1832	0.1755	0.4382	0.4143
TiRex	0.0521	0.0256	0.0374	0.0184	0.0517	0.0358	0.0853	0.0502	0.0066	0.0030	<u>0.0097</u>	0.0063
MOMENT-Large	0.0636	0.0333	0.0417	0.0221	0.0937	0.0741	0.1133	0.0710	0.0105	0.0047	0.0161	0.0093
MOMENT-Base	0.0668	0.0352	0.0451	0.0246	0.0946	0.0738	0.1177	0.0730	0.0096	0.0043	0.0170	0.0096
MOMENT-Small	0.0657	0.0335	0.0436	0.0234	0.0892	0.0705	0.1189	0.0738	0.0107	0.0049	0.0166	0.0091
MoiraiMoE-Base	0.0744	0.0411	0.0501	0.0237	0.1237	0.0844	0.1118	0.0706	0.0140	0.0052	0.0138	0.0109
MoiraiMoE-Small	0.0775	0.0428	0.0479	0.0227	0.1265	0.0919	0.1155	0.0716	0.0129	0.0049	0.0139	0.0112
Moirai2.0-Small	0.0526	0.0254	0.0395	0.0194	0.0785	0.0407	0.0867	0.0548	<u>0.0031</u>	<u>0.0012</u>	0.0123	0.0097
TimesFM 1.0	0.0590	0.0270	0.0439	0.0212	0.0944	0.0719	0.0854	0.0538	<u>0.0039</u>	<u>0.0014</u>	0.0121	0.0083
TimesFM 2.0	0.0623	0.0301	0.0389	0.0192	0.1004	0.0762	0.0868	0.0559	0.0032	<u>0.0012</u>	0.0117	0.0072
TimesFM 2.5	0.0508	0.0235	0.0364	0.0181	0.0934	0.0681	0.0845	0.0530	0.0013	0.0005	0.0125	0.0093
TimeMoE-Large	0.0123	0.0097	<u>0.0171</u>	<u>0.0101</u>	<u>0.0299</u>	<u>0.0187</u>	0.0170	<u>0.0107</u>	0.0283	0.0241	0.0152	0.0117
TimeMoE-Base	0.0160	0.0116	0.0237	0.0167	0.0360	0.0225	<u>0.0176</u>	0.0117	0.0314	0.0252	0.0225	0.0172
Chronos2-Base	0.0570	0.0250	0.0345	0.0164	0.0636	0.0354	<u>0.0869</u>	0.0531	0.0046	0.0017	0.0117	<u>0.0052</u>
Chronos2-Small	0.0562	0.0251	0.0345	0.0164	0.0851	0.0570	0.0858	0.0529	0.0052	0.0019	0.0121	<u>0.0054</u>
Sundial-Base	0.0525	0.0254	0.0386	0.0194	0.0999	0.0526	0.0819	0.0526	0.0080	0.0065	0.0122	0.0077
Aurora	0.0647	0.0370	0.0495	0.0256	0.1206	0.0926	0.1044	0.0662	0.0237	0.0089	0.0139	0.0067
MIRA-Large	<u>0.0136</u>	<u>0.0100</u>	0.0203	0.0153	0.0456	0.0359	0.0258	0.0205	0.0596	0.0519	0.0335	0.0300
SOTER	0.0152	0.0101	0.0109	0.0054	0.0217	0.0112	0.0204	0.0103	0.0103	0.0071	0.0064	0.0051

Table 12: Complete continuous-time imputation results at a 25% missing rate. Each entry reports the mean and standard deviation over three runs with different random seeds in the normalized space. The mean values are used to construct Figure 2. Bold and underlined means denote the best and second-best results in each column, respectively.

Method	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)		IEEE PPG (MC)		MMASH (MC)		WDD (MC)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
TimeMoE-Large	0.1785 $\pm$ 0.0018	0.1615 $\pm$ 0.0014	0.1748 $\pm$ 0.0013	0.1603 $\pm$ 0.0006	0.1801 $\pm$ 0.0035	0.1598 $\pm$ 0.0017	0.1891 $\pm$ 0.0021	0.1649 $\pm$ 0.0008	0.1347 $\pm$ 0.0024	0.1268 $\pm$ 0.0013	0.1785 $\pm$ 0.0018	0.1614 $\pm$ 0.0013
Moirai2.0-Small	0.0510 $\pm$ 0.0122	0.0281 $\pm$ 0.0054	0.1083 $\pm$ 0.0004	0.0724 $\pm$ 0.0002	0.0819 $\pm$ 0.0016	0.0474 $\pm$ 0.0012	0.1214 $\pm$ 0.0026	0.0797 $\pm$ 0.0016	0.0292 $\pm$ 0.0005	0.0175 $\pm$ 0.0004	0.0460 $\pm$ 0.0022	0.0261 $\pm$ 0.0014
MOMENT-Large	0.0535 $\pm$ 0.0043	0.0322 $\pm$ 0.0006	0.1148 $\pm$ 0.0123	0.1003 $\pm$ 0.0179	0.1018 $\pm$ 0.0047	0.0798 $\pm$ 0.0073	0.1698 $\pm$ 0.0167	0.1425 $\pm$ 0.0177	0.0436 $\pm$ 0.0023	0.0340 $\pm$ 0.0018	0.1117 $\pm$ 0.0071	0.0943 $\pm$ 0.0052
Chronos2-Base	0.0466 $\pm$ 0.0101	0.0220 $\pm$ 0.0038	0.0578 $\pm$ 0.0015	0.0251 $\pm$ 0.0004	0.0767 $\pm$ 0.0052	0.0313 $\pm$ 0.0023	0.0947 $\pm$ 0.0057	0.0503 $\pm$ 0.0036	0.0252 $\pm$ 0.0008	0.0179 $\pm$ 0.0007	0.0516 $\pm$ 0.0038	0.0149 $\pm$ 0.0012
TiRex	0.0267 $\pm$ 0.0016	0.0232 $\pm$ 0.0009	0.0355 $\pm$ 0.0016	0.0154 $\pm$ 0.0009	0.0554 $\pm$ 0.0057	0.0268 $\pm$ 0.0053	0.0690 $\pm$ 0.0063	0.0381 $\pm$ 0.0039	0.0125 $\pm$ 0.0015	0.0025 $\pm$ 0.0011	0.0167 $\pm$ 0.0026	0.0142 $\pm$ 0.0017
Sundial-Base	0.0549 $\pm$ 0.0044	0.0324 $\pm$ 0.0020	0.1063 $\pm$ 0.0003	0.0840 $\pm$ 0.0009	0.1131 $\pm$ 0.0028	0.0825 $\pm$ 0.0015	0.1310 $\pm$ 0.0021	0.0993 $\pm$ 0.0008	0.0419 $\pm$ 0.0017	0.0308 $\pm$ 0.0009	0.0933 $\pm$ 0.0016	0.0701 $\pm$ 0.0010
Aurora	0.0860 $\pm$ 0.0057	0.0537 $\pm$ 0.0032	0.1701 $\pm$ 0.0003	0.1528 $\pm$ 0.0003	0.1824 $\pm$ 0.0025	0.1511 $\pm$ 0.0015	0.1967 $\pm$ 0.0019	0.1681 $\pm$ 0.0011	0.0689 $\pm$ 0.0015	0.0576 $\pm$ 0.0009	0.1555 $\pm$ 0.0023	0.1345 $\pm$ 0.0011
MIRA-Large	0.1101 $\pm$ 0.0035	0.0949 $\pm$ 0.0024	0.1782 $\pm$ 0.0005	0.1627 $\pm$ 0.0006	0.1926 $\pm$ 0.0018	0.1710 $\pm$ 0.0015	0.1999 $\pm$ 0.0013	0.1766 $\pm$ 0.0005	0.1542 $\pm$ 0.0017	0.1473 $\pm$ 0.0016	0.1895 $\pm$ 0.0011	0.1735 $\pm$ 0.0009
SOTER	0.0253 $\pm$ 0.0007	0.0205 $\pm$ 0.0003	0.0183 $\pm$ 0.0002	0.0083 $\pm$ 0.0001	0.0303 $\pm$ 0.0009	0.0142 $\pm$ 0.0005	0.0391 $\pm$ 0.0005	0.0182 $\pm$ 0.0004	0.0204 $\pm$ 0.0003	0.0132 $\pm$ 0.0003	0.0152 $\pm$ 0.0004	0.0091 $\pm$ 0.0001

Table 13: Complete continuous-time imputation results at a 50% missing rate. Reporting conventions follow Table 12.

Method	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)		IEEE PPG (MC)		MMASH (MC)		WDD (MC)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
TimeMoE-Large	0.1877 $\pm$ 0.0027	0.1485 $\pm$ 0.0047	0.2738 $\pm$ 0.0006	0.2645 $\pm$ 0.0011	0.2761 $\pm$ 0.0116	0.2529 $\pm$ 0.0122	0.2908 $\pm$ 0.0010	0.2699 $\pm$ 0.0013	0.1325 $\pm$ 0.0055	0.1227 $\pm$ 0.0047	0.2521 $\pm$ 0.0016	0.2371 $\pm$ 0.0019
Moirai2.0-Small	0.1519 $\pm$ 0.0459	0.1122 $\pm$ 0.0272	0.2083 $\pm$ 0.0081	0.1657 $\pm$ 0.0079	0.1739 $\pm$ 0.0104	0.1177 $\pm$ 0.0097	0.2574 $\pm$ 0.0055	0.2086 $\pm$ 0.0055	0.0615 $\pm$ 0.0020	0.0434 $\pm$ 0.0039	0.0766 $\pm$ 0.0082	0.0500 $\pm$ 0.0041
MOMENT-Large	0.0928 $\pm$ 0.0046	0.0627 $\pm$ 0.0020	0.2331 $\pm$ 0.0195	0.2144 $\pm$ 0.0267	0.1958 $\pm$ 0.0246	0.1638 $\pm$ 0.0258	0.2729 $\pm$ 0.0206	0.2465 $\pm$ 0.0223	0.0781 $\pm$ 0.0043	0.0628 $\pm$ 0.0029	0.2742 $\pm$ 0.0153	0.2490 $\pm$ 0.0159
Chronos2-Base	0.1357 $\pm$ 0.0024	0.0982 $\pm$ 0.0026	0.3028 $\pm$ 0.0038	0.2811 $\pm$ 0.0035	0.3112 $\pm$ 0.0133	0.2690 $\pm$ 0.0114	0.3430 $\pm$ 0.0078	0.3078 $\pm$ 0.0063	0.1214 $\pm$ 0.0041	0.1036 $\pm$ 0.0032	0.2667 $\pm$ 0.0085	0.2287 $\pm$ 0.0053
TiRex	0.0462 $\pm$ 0.0029	0.0240 $\pm$ 0.0022	0.0442 $\pm$ 0.0020	0.0198 $\pm$ 0.0008	0.0800 $\pm$ 0.0027	0.0361 $\pm$ 0.0030	0.0993 $\pm$ 0.0033	0.0560 $\pm$ 0.0027	0.0166 $\pm$ 0.0011	0.0056 $\pm$ 0.0010	0.0297 $\pm$ 0.0007	0.0202 $\pm$ 0.0012
Sundial-Base	0.0997 $\pm$ 0.0185	0.0761 $\pm$ 0.0106	0.2737 $\pm$ 0.0007	0.2594 $\pm$ 0.0001	0.2806 $\pm$ 0.0057	0.2471 $\pm$ 0.0062	0.3007 $\pm$ 0.0035	0.2751 $\pm$ 0.0024	0.1078 $\pm$ 0.0032	0.0969 $\pm$ 0.0035	0.2412 $\pm$ 0.0041	0.2191 $\pm$ 0.0032
Aurora	0.1171 $\pm$ 0.0075	0.0839 $\pm$ 0.0074	0.2877 $\pm$ 0.0025	0.2765 $\pm$ 0.0019	0.2949 $\pm$ 0.0054	0.2628 $\pm$ 0.0065	0.3178 $\pm$ 0.0041	0.2953 $\pm$ 0.0028	0.1152 $\pm$ 0.0020	0.1035 $\pm$ 0.0021	0.2574 $\pm$ 0.0036	0.2387 $\pm$ 0.0031
MIRA-Large	0.1249 $\pm$ 0.0345	0.1008 $\pm$ 0.0216	0.2714 $\pm$ 0.0019	0.2609 $\pm$ 0.0015	0.2754 $\pm$ 0.0044	0.2498 $\pm$ 0.0052	0.2945 $\pm$ 0.0026	0.2749 $\pm$ 0.0010	0.1748 $\pm$ 0.0033	0.1550 $\pm$ 0.0037	0.2532 $\pm$ 0.0042	0.2383 $\pm$ 0.0034
SOTER	0.0280 $\pm$ 0.0003	0.0177 $\pm$ 0.0010	0.0307 $\pm$ 0.0009	0.0177 $\pm$ 0.0006	0.0459 $\pm$ 0.0043	0.0256 $\pm$ 0.0019	0.0520 $\pm$ 0.0009	0.0295 $\pm$ 0.0016	0.0262 $\pm$ 0.0004	0.0192 $\pm$ 0.0003	0.0270 $\pm$ 0.0010	0.0195 $\pm$ 0.0003

Table 14: Complete continuous-time imputation results at a 75% missing rate. Reporting conventions follow Table 12.

Method	HeartRate (SC)		MIT-BIH (MC)		ScientiSST MOVE (MC)		IEEE PPG (MC)		MMASH (MC)		WDD (MC)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
TimeMoE-Large	0.2114 $\pm$ 0.0063	0.1679 $\pm$ 0.0056	0.3786 $\pm$ 0.0020	0.3721 $\pm$ 0.0019	0.3787 $\pm$ 0.0076	0.3492 $\pm$ 0.0088	0.4027 $\pm$ 0.0045	0.3855 $\pm$ 0.0038	0.1649 $\pm$ 0.0047	0.1498 $\pm$ 0.0064	0.3388 $\pm$ 0.0014	0.3206 $\pm$ 0.0020
Moirai2.0-Small	0.1808 $\pm$ 0.0398	0.1490 $\pm$ 0.0333	0.2885 $\pm$ 0.0171	0.2471 $\pm$ 0.0185	0.2338 $\pm$ 0.0117	0.1713 $\pm$ 0.0112	0.3516 $\pm$ 0.0038	0.3072 $\pm$ 0.0061	0.0883 $\pm$ 0.0039	0.0691 $\pm$ 0.0021	0.1378 $\pm$ 0.0063	0.0881 $\pm$ 0.0087
MOMENT-Large	0.1328 $\pm$ 0.0179	0.1000 $\pm$ 0.0113	0.3510 $\pm$ 0.0302	0.3430 $\pm$ 0.0311	0.3269 $\pm$ 0.0168	0.2945 $\pm$ 0.0157	0.3911 $\pm$ 0.0108	0.3707 $\pm$ 0.0119	0.1088 $\pm$ 0.0045	0.0883 $\pm$ 0.0044	0.3594 $\pm$ 0.0027	0.3383 $\pm$ 0.0024
Chronos2-Base	0.2047 $\pm$ 0.0651	0.1617 $\pm$ 0.0425	0.4850 $\pm$ 0.0014	0.4809 $\pm$ 0.0014	0.4816 $\pm$ 0.0101	0.4491 $\pm$ 0.0123	0.5100 $\pm$ 0.0050	0.4959 $\pm$ 0.0046	0.1895 $\pm$ 0.0050	0.1800 $\pm$ 0.0076	0.4280 $\pm$ 0.0050	0.4075 $\pm$ 0.0049
TiRex	0.0714 $\pm$ 0.0059	0.0392 $\pm$ 0.0032	0.1603 $\pm$ 0.0036	0.1122 $\pm$ 0.0063	0.1722 $\pm$ 0.0083	0.1122 $\pm$ 0.0063	0.1897 $\pm$ 0.0059	0.1093 $\pm$ 0.0044	0.0789 $\pm$ 0.0025	0.0509 $\pm$ 0.0035	0.1463 $\pm$ 0.0094	0.1126 $\pm$ 0.0041
Sundial-Base	0.1076 $\pm$ 0.0065	0.0974 $\pm$ 0.0029	0.4105 $\pm$ 0.0101	0.4052 $\pm$ 0.0110	0.4083 $\pm$ 0.0082	0.3823 $\pm$ 0.0059	0.4499 $\pm$ 0.0047	0.4335 $\pm$ 0.0062	0.1955 $\pm$ 0.0037	0.1700 $\pm$ 0.0014	0.3601 $\pm$ 0.0115	0.3503 $\pm$ 0.0136
Aurora	0.1642 $\pm$ 0.0254	0.1160 $\pm$ 0.0124	0.4030 $\pm$ 0.0021	0.3966 $\pm$ 0.0027	0.4069 $\pm$ 0.0186	0.3759 $\pm$ 0.0199	0.4322 $\pm$ 0.0008	0.4149 $\pm$ 0.0026	0.1881 $\pm$ 0.0072	0.1649 $\pm$ 0.0087	0.3605 $\pm$ 0.0014	0.3405 $\pm$ 0.0027
MIRA-Large	0.1556 $\pm$ 0.0298	0.1226 $\pm$ 0.0230	0.3720 $\pm$ 0.0026	0.3648 $\pm$ 0.0033	0.3756 $\pm$ 0.0173	0.3464 $\pm$ 0.0181	0.3962 $\pm$ 0.0017	0.3782 $\pm$ 0.0020	0.1991 $\pm$ 0.0046	0.1507 $\pm$ 0.0058	0.3335 $\pm$ 0.0041	0.3161 $\pm$ 0.0038
SOTER	0.0546 $\pm$ 0.0071	0.0317 $\pm$ 0.0020	0.1502 $\pm$ 0.0047	0.0860 $\pm$ 0.0041	0.1644 $\pm$ 0.0050	0.0915 $\pm$ 0.0033	0.1684 $\pm$ 0.0059	0.0984 $\pm$ 0.0039	0.0541 $\pm$ 0.0029	0.0408 $\pm$ 0.0019	0.1282 $\pm$ 0.0042	0.0730 $\pm$ 0.0025

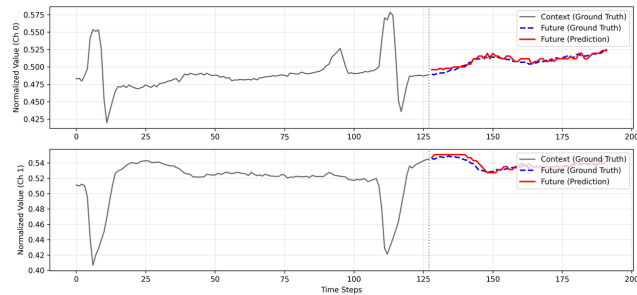


Figure 6: Zero-shot forecasting example on a randomly selected two-channel MIT-BIH sample under the 128/64 input/output setting. The gray curves denote the observed context, the blue dashed curves denote the ground-truth future, and the red curves denote SOTER’s predictions. The vertical dotted line marks the forecasting origin.

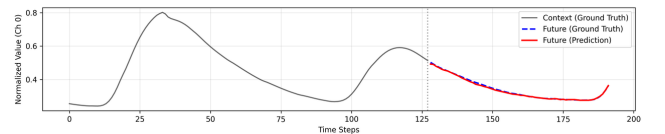


Figure 7: Zero-shot forecasting example on a randomly selected single-channel HeartRate sample under the 128/64 input/output setting. The gray curve denotes the observed context, the blue dashed curve denotes the ground-truth future, and the red curve denotes SOTER’s prediction. The vertical dotted line marks the forecasting origin.

H Zero-Shot Forecasting Visualizations

To complement the quantitative results in Appendix F, we visualize representative zero-shot forecasts produced by SOTER under the longest input/output window, 128/64. The samples are randomly

Table 15: Label-efficient frozen linear probing on ScientiSST MOVE under varying supervision budgets. Results are reported as mean $\pm$ std over three runs. Best scores are highlighted by bold style and second-best are underlined.

Methods	Macro-F1						Macro-AUROC					
	1%	5%	10%	25%	50%	100%	1%	5%	10%	25%	50%	100%
TF-C	31.68 $\pm$ 0.92	33.65 $\pm$ 0.55	34.68 $\pm$ 0.51	37.26 $\pm$ 0.57	43.51 $\pm$ 0.60	47.79 $\pm$ 2.35	83.24 $\pm$ 0.43	<u>86.33</u> $\pm$ 0.69	87.81 $\pm$ 0.39	88.51 $\pm$ 0.52	90.21 $\pm$ 0.41	91.89 $\pm$ 2.02
MOMENT-Large	30.87 $\pm$ 1.56	34.57 $\pm$ 0.75	<u>42.72</u> $\pm$ 0.31	44.99 $\pm$ 0.67	47.91 $\pm$ 0.79	49.60 $\pm$ 0.55	81.41 $\pm$ 0.44	83.88 $\pm$ 0.31	86.61 $\pm$ 0.36	87.78 $\pm$ 0.48	88.48 $\pm$ 0.70	88.55 $\pm$ 0.64
NormWear	<u>35.48</u> $\pm$ 0.81	<u>38.91</u> $\pm$ 0.49	40.79 $\pm$ 0.68	42.66 $\pm$ 0.38	42.90 $\pm$ 0.99	43.03 $\pm$ 0.16	<u>84.19</u> $\pm$ 0.55	85.92 $\pm$ 0.34	88.36 $\pm$ 0.38	<u>91.21</u> $\pm$ 0.40	<u>90.64</u> $\pm$ 1.14	<u>91.92</u> $\pm$ 0.98
MIRA-Large	32.47 $\pm$ 0.99	36.80 $\pm$ 0.67	41.50 $\pm$ 0.51	<u>47.32</u> $\pm$ 0.87	<u>48.70</u> $\pm$ 0.86	<u>50.09</u> $\pm$ 1.22	83.86 $\pm$ 0.61	85.38 $\pm$ 0.45	<u>88.77</u> $\pm$ 0.35	91.02 $\pm$ 0.84	89.69 $\pm$ 0.78	90.07 $\pm$ 0.81
SOTER	40.48 $\pm$ 0.83	41.69 $\pm$ 0.45	44.87 $\pm$ 0.46	48.01 $\pm$ 0.88	50.42 $\pm$ 0.54	50.90 $\pm$ 0.36	89.03 $\pm$ 0.56	91.38 $\pm$ 0.45	93.41 $\pm$ 0.52	94.89 $\pm$ 0.43	94.95 $\pm$ 0.24	95.62 $\pm$ 0.12

selected from MIT-BIH and HeartRate to illustrate forecasting behavior on multi-channel and single-channel physiological signals, respectively. In both figures, the gray curve denotes the observed context, the blue dashed curve denotes the ground-truth future, the red curve denotes SOTER’s prediction, and the vertical dotted line marks the forecasting origin. Figure 6 presents a two-channel MIT-BIH example, whereas Figure 7 presents a single-channel HeartRate example.

I Label-Efficient Frozen-Representation Evaluation

To examine the label efficiency of the frozen representations, we evaluate linear probing on ScientiSST MOVE using 1%, 5%, 10%, 25%, 50%, and 100% of the labeled training samples, while keeping every pre-trained backbone fixed. As shown in Table 15, SOTER achieves the best Macro-F1 and Macro-AUROC at every supervision level. The advantage is particularly clear in the low-label regime: with only 1% of the labeled data, SOTER obtains a Macro-F1 of 40.48

and a Macro-AUROC of 89.03, exceeding the strongest competing results by 5.00 and 4.84 percentage points, respectively. With 25% of the labels, SOTER already reaches 48.01 Macro-F1 and 94.89 Macro-AUROC, compared with 50.90 and 95.62 under full supervision. These results demonstrate that, on ScientiSST MOVE, SOTER’s frozen representations remain effective when downstream labels are limited.

J LLM Usage Declaration

A large language model (LLM) was used as an auxiliary tool during the preparation of this work, primarily for code-level assistance and English-language refinement. The research problem, methodological design, dataset selection and processing, experimental protocol, mathematical analysis, interpretation of results, and scientific conclusions were developed by the human authors. Any LLM-assisted code or text was reviewed, revised, and verified by the authors, who take full responsibility for the correctness and integrity of the submitted work.